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PAPER

DeepLNP: role-aware dual-path learning for lipid nanoparticle delivery efficacy modelling and within-screen ranking

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Abstract

Lipid nanoparticle (LNP) delivery depends on lipid chemistry, composition, biological target and experimental context. We present DeepLNP, a role-aware dual-path model representing four lipid roles, molar composition and context. A shared molecular-graph pathway predicts formulation-level endpoints, while a gated ionizable-lipid Morgan-fingerprint residual supports within-screen ranking. Masked heads handle missing physicochemical outcomes and censored toxicity observations. We evaluated 13,605 records, including 12,692 screen-standardized delivery measurements from 38 screens, using five group-disjoint folds. DeepLNP achieved RMSE of 1.003 ± 0.003 and mean within-screen Spearman correlation of 0.222 ± 0.067 . The ranking residual increased Spearman correlation from 0.102 for the graph-only model to 0.222 without degrading point error. At inspection budgets from 5% to 20%, top-set recovery increased from 0.091 to 0.305 and normalized best-candidate regret fell from 0.278 to 0.100. Matched conventional baselines showed complementary strengths for the two objectives. Metadata Extra Trees reduced RMSE to 0.988 ± 0.024 , while ionizable-lipid Morgan Extra Trees increased Spearman correlation to 0.245 ± 0.049 . The validation-selected hybrid reached 0.988 ± 0.021 RMSE and 0.248 ± 0.054 Spearman correlation. Bootstrap intervals did not establish its superiority over the strongest standalone comparator. DeepLNP provides a unified representation of formulation and context with separate outputs for point estimation and candidate ranking. This design supports objective-specific formulation prioritization across held-out experimental screens.

Key words: lipid nanoparticles, deep learning, drug delivery, within-screen ranking, molecular representation

Introduction

Lipid nanoparticles (LNPs) have emerged as a versatile platform for delivering messenger RNA, small interfering RNA and genome-editing cargo [1, 2]. A canonical nucleic-acid LNP typically comprises an ionizable lipid, a helper phospholipid, a sterol and a PEG-lipid, whose chemical identities and relative proportions jointly influence particle formation, stability and delivery behaviour [3, 4]. Beyond lipid composition, manufacturing conditions can modulate intramuscular expression and intracellular delivery [5, 6], while ionizable-lipid chemistry strongly affects endosomal escape and tolerability [7]. High-throughput screening has further enabled the identification of LNPs with cell- and tissue-dependent delivery profiles [8, 9], and additional targeting lipids can redirect biodistribution among organs [10]. LNP delivery should therefore be viewed as an outcome of a multicomponent formulation embedded within a biological and experimental context.

Machine learning has increasingly been used to accelerate LNP design and ionizable-lipid discovery. Graph-based models,

combinatorial chemistry and learned structure–activity relationships have enabled data-driven exploration of chemical space [11, 12, 13], whereas SMILES-based embeddings, balanced transformers and multimodal molecular representations provide alternative encodings of lipid structure [14, 15, 16]. More recent approaches incorporate cargo, assay or biological-target information [17], while formulation-aware models additionally encode component identities, molar composition, N/P ratio and process variables [18, 19, 20, 21, 22, 23, 24, 25, 26]. Collectively, these studies demonstrate the value of combining molecular and contextual information for LNP prediction. However, several practical aspects of formulation screening remain only partially addressed within a unified framework. LNP components occupy distinct physicochemical roles and are therefore not naturally interchangeable; formulation datasets often contain incompletely observed physicochemical or safety-related endpoints; and the ultimate screening decision depends on prioritizing candidates within an experimental campaign rather than solely minimizing pooled regression error.

A further challenge arises from the heterogeneity of publicly available LNP data. Experimental studies differ substantially

in formulation space, manufacturing conditions, assay design, biological targets and measurement protocols [27, 28, 29]. Consequently, random row-wise partitioning can place observations from the same experimental campaign in both training and test sets, potentially inflating apparent generalization. Similar partition-dependent effects are well recognized in molecular property prediction [30, 31], and recent benchmarking of LNP transfection models has shown that data curation, partitioning strategy and conventional machine-learning baselines can materially alter conclusions about the value of more complex molecular representations [32]. Evaluation of LNP models should therefore distinguish interpolation within familiar experimental settings from generalization to entirely held-out screens and should assess point prediction separately from screen-level candidate prioritization.

Here, we introduce DeepLNP, a role-aware, target-conditioned deep learning framework designed to separate formulation-level point estimation from within-screen candidate ranking (Figure 1). DeepLNP uses a shared molecular-graph pathway to encode the four canonical lipid roles, their molar fractions and contextual information for endpoint prediction. A complementary evidence-aligned pathway encodes ionizable-lipid Morgan fingerprints together with formulation and biological context and contributes a gated residual exclusively to the ranking score. This dual-path design allows structural graph representations to support formulation-level prediction while retaining substructure information that is informative for relative candidate ordering within a screening campaign. Masked heteroscedastic heads provide a common interface for partially observed physicochemical endpoints, while bounded toxicity observations are represented using a censored likelihood. We evaluate DeepLNP using five endpoint-balanced, group-disjoint folds spanning 38 held-out experimental screens and report formulation-level prediction error independently from within-screen ranking and finite-budget candidate-selection metrics. This evaluation tests whether explicitly separating point estimation from candidate prioritization provides a useful modeling strategy under experimental-group shift, while maintaining a unified role-aware representation of formulation, target and process context.

Materials and methods

Problem formulation and data support

For formulation n and lipid role $i \in \{\text{ion, helper, sterol, PEG}\}$, DeepLNP receives a molecular structure S_{ni} when resolved, a reported molar fraction r_{ni} , a structure-validity mask m_{ni}^{str} and a positive-fraction activity mask m_{ni}^{act} . These role-specific inputs are combined with process variables $\mathbf{p}_n$, assay context $\mathbf{c}_n$ and biological-target context $\mathbf{t}_n$:

$$\mathcal{X}_n = \left(\{S_{ni}, r_{ni}, m_{ni}^{\text{str}}, m_{ni}^{\text{act}}\}_{i=1}^4, \mathbf{p}_n, \mathbf{c}_n, \mathbf{t}_n \right). \quad (1)$$

The outputs are means and log variances for delivery efficiency, particle size, polydispersity index (PDI), zeta potential, encapsulation efficiency and toxicity, together with an efficacy-ranking score. Endpoint masks restrict each loss to rows with an observed or bounded response.

The project loader reconstructed 13,605 records from two integrated resources: 12,692 high-throughput delivery records and 913 literature-derived formulation records. Delivery efficiency was available for 12,692 rows, particle size for 905, PDI for 649, zeta potential for 265, encapsulation efficiency for 570 and

numeric or bounded toxicity for 99. Delivery responses span 38 context-specific screens. For upstream response a_n in screen $g(n)$, the released target is

$$y_n = \frac{a_n - \bar{a}_{g(n)}}{s_{g(n)}}, \quad s_g^2 = \frac{1}{|g|} \sum_{m \in g} (a_m - \bar{a}_g)^2. \quad (2)$$

The released delivery labels have mean -0.0055 , standard deviation 1.0026 and range $[-3.157, 7.224]$. An RMSE near one therefore represents approximately one upstream within-screen standard deviation, not an error in percentage transfection. The auxiliary endpoints do not overlap delivery row-wise. Although 732 records contain resolved structures for all four lipid roles, none carries a delivery label. The delivery set contains 9,194 unique ionizable-lipid graphs and four helper-lipid graphs; 12,093 rows use the single cholesterol identity specified by the source protocol, and the reported PEG2000 product label does not resolve to a unique molecular graph. The data therefore support composition-aware prediction but not delivery-side learning of sterol or PEG structural variation. Full support counts and overlap matrices are reported in the Supplementary Material.

DeepLNP architecture

Role-aware molecular and composition encoding

A shared two-layer message-passing graph encoder maps 39-dimensional atom features and four bond-type indicators to a 64-dimensional representation. Sharing the chemical encoder controls parameter count while a learned role embedding distinguishes the four functional positions. For each role, the graph representation is combined with a molar-fraction encoding and six deterministic chemical descriptors:

$$\mathbf{h}_{ni} = G(S_{ni}) + \mathbf{e}_i + \phi(r_{ni}) + D(\mathbf{d}_{ni}). \quad (3)$$

Two four-head self-attention blocks transform the role tokens. Molar fractions enter both the role-token encoder and the global formulation branch, allowing local component interactions and global process context to use the same composition. Missing structures are zero-encoded and retained with validity metadata. In the selected configuration, the validity bit is not added to the token as an explicit predictive feature; attention activity follows the reported positive-fraction mask. The dummy graph and zero-descriptor pattern preserve an implicit missingness signal, so the learned representation can remain associated with source-linked structure availability.

Target- and process-conditioned fusion

An 11-class biological-target query cross-attends to the role tokens. A separate branch encodes 14 formulation and process quantities together with 14 observation indicators, thereby distinguishing an unreported value from a physical zero. Assay categories describe cargo, model system and route; explicit source, paper and screen identifiers are fixed to zero or excluded. The conditioned molecular representation and the global formulation branch are projected to a common 64-dimensional space and combined by a three-expert gated fusion module. The selected primary model uses deterministic two-dimensional descriptors, but conformer generation is disabled, so its six shape channels contain no three-dimensional evidence.

Masked multi-endpoint and censored prediction

For measured endpoint q , DeepLNP predicts mean μ_{nq} and log variance ℓ_{nq} and minimizes a masked Gaussian negative log

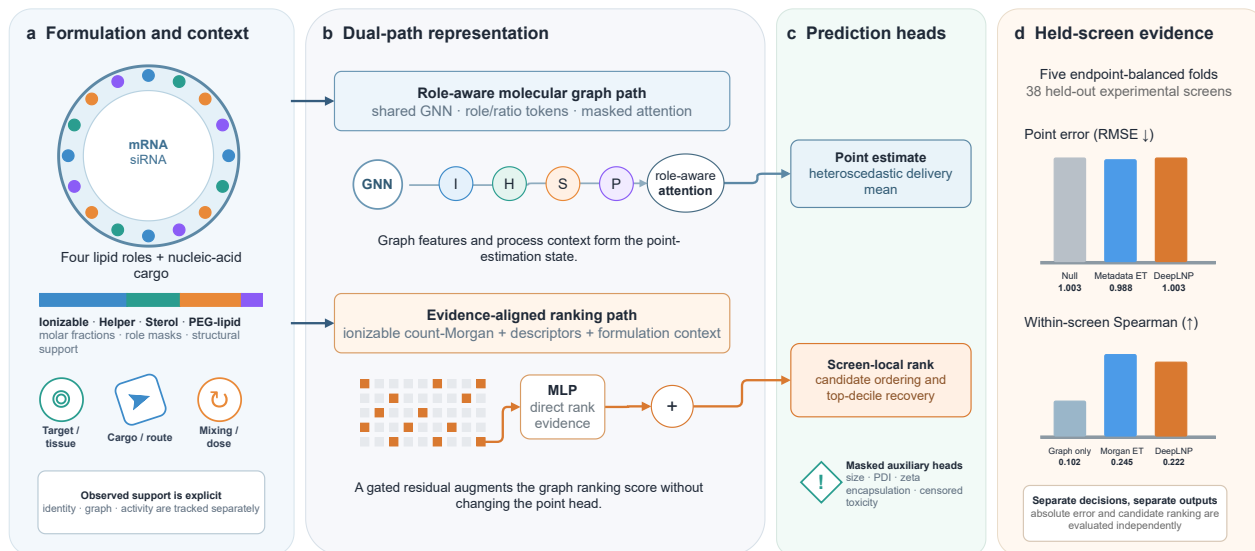


Fig. 1. DeepLNP dual-path architecture. **a**, Role-resolved inputs combine four lipid slots, molar composition and cargo, biological and manufacturing context while recording activity, identity and graph support separately. **b**, The graph pathway uses a shared molecular encoder, role-ratio tokens and masked attention for endpoint estimation. The evidence-aligned pathway encodes an ionizable-lipid count-Morgan vector, deterministic descriptors and formulation/context features and contributes a gated residual only to the ranking score. **c**, Separate delivery-mean and screen-ranking outputs accompany masked physicochemical and censored-toxicity heads. **d**, Five-fold held-screen evidence compares the null, matched tree baselines and DeepLNP.

Alt text: Four aligned panels show the four-component LNP and contextual inputs, a graph encoder beside a direct Morgan ranking pathway, separate point and rank outputs with auxiliary physical and toxicity heads, and compact held-screen RMSE and Spearman comparisons.

likelihood over observed set Ω_q :

$$\mathcal{L}_q = \frac{1}{|\Omega_q|} \sum_{n \in \Omega_q} \left\{ \frac{(y_{nq} - \mu_{nq})^2}{2 \exp(\ell_{nq})} + \frac{\ell_{nq}}{2} \right\}. \quad (4)$$

The 99 retained toxicity values are upper bounds on risk derived from explicit viability statements. Their head uses the corresponding censored Gaussian probability rather than treating a bound as an exact value [33]. An auxiliary target classifier is applied to the unconditional molecular representation, before the target query is introduced, and contributes cross-entropy only for rows with a non-missing target class. This placement prevents the target input from being copied directly to its auxiliary label. The heads form a common partially observed endpoint interface across source-blocked tasks, without paired delivery-physicochemical supervision.

Dual-path screen-aware ranking

The graph pathway first produces a delivery mean and a graph-derived ranking score

$$r_n^{\text{graph}} = \text{stopgrad}(\mu_n) + \sigma(\gamma) a_\psi(\mathbf{z}_n), \quad (5)$$

where $\mathbf{z}_n$ is the fused formulation representation, a_ψ is a learned adapter, σ is the sigmoid function and γ is a learned scalar gate. An evidence-aligned vector $\mathbf{u}_n$ concatenates the log-count ionizable-lipid Morgan fingerprint, deterministic descriptors, formulation/process values, assay and target context, and role masks. A two-layer perceptron b_ω supplies a second gated residual:

$$r_n = r_n^{\text{graph}} + \sigma(\eta) b_\omega(\mathbf{u}_n). \quad (6)$$

The direct pathway changes the ranking score but not μ_n . Delivery pairs are sampled only from same-screen subsets within a

minibatch. The selected model combines pairwise ranking with a screen-centred objective; a low-weight ListMLE term was evaluated separately [34, 35, 36]. The selected training objective is

$$\mathcal{L} = \sum_q \lambda_q \mathcal{L}_q + \lambda_{\text{pair}} \mathcal{L}_{\text{pair}} + \lambda_{\text{centre}} \mathcal{L}_{\text{centre}} + \lambda_{\text{target}} \mathcal{L}_{\text{target}}, \quad (7)$$

where endpoint terms are included only when labels are present. The ListMLE term is excluded from this primary objective and appears only in the ablation study.

Training, baselines and model selection

The selected DeepLNP configuration contains 529,415 trainable parameters, uses AdamW with learning rate 5×10^{-4} and weight decay 10^{-4} , mixed precision, a maximum of six epochs and patience two. Screen-paired batches sample four groups and four rows per group when sufficient candidates are available. A validation-only multi-metric checkpoint score is dominated by delivery RMSE and includes smaller correlation and ranking terms. Test groups do not contribute to preprocessing, checkpoint selection or objective choice.

The matched baseline panel contains a zero predictor, the training-set mean, metadata-only Extra Trees, ionizable-Morgan-plus-metadata Extra Trees and role-wise-Morgan-plus-metadata Extra Trees. Metadata comprises the same molar fractions, formulation/process values and observation indicators, target and assay context, and role-validity and activity masks available to DeepLNP. Explicit source, publication, screen and split identifiers are excluded. Extra Trees minimum leaf size is selected independently in each fold using validation groups. The hybrid combines the validation-selected metadata tree for the point output and ionizable-Morgan tree for the ranking output

with DeepLNP, selecting separate weights from $\{0, 0.1, \dots, 1\}$ by validation RMSE and validation mean screen Spearman. Ranking scores are standardized within each candidate batch before combination.

Group-disjoint evaluation and statistical analysis

An experimental group used the most specific available identifier: delivery screen, DOI, paper title or source only when no finer identifier existed. Group assignment was performed on the complete corpus before endpoint filtering. The primary evaluation used five endpoint-balanced folds constructed from group identity, group size and endpoint availability, never endpoint values. Each fold contained 2,721 corpus rows and 2,532–2,548 delivery observations; every delivery screen served once as validation and once as test. Three frozen group-shuffle partitions were used only for objective and input ablations.

Point prediction was evaluated primarily by RMSE. Screening metrics were computed inside each held-out screen before aggregation: mean screen Spearman, top-decile hit recovery (Hit@10%) and range-normalized regret. For predicted top- k set $\hat{T}_{g,k}$,

$$R_{g,k}^{\text{norm}} = \frac{\max_{i \in g} y_i - \max_{i \in \hat{T}_{g,k}} y_i}{\max_{i \in g} y_i - \min_{i \in g} y_i} \quad (8)$$

NDCG, pair accuracy, MAE, R^2 , Pearson and pooled Spearman were secondary. Confidence intervals resampled complete screens. Robustness analyses compared row-random with group-disjoint partitions, equal-screen with row-count weighting, exclusion of the largest screen and prespecified size strata.

Practical screening utility was evaluated from the frozen out-of-fold ranking scores at budgets of 5%, 10% and 20%, with $k = \lceil qn_g \rceil$ candidates inspected in a screen of size n_g . Context summaries averaged complete screens within delivery target, administration route or cargo strata. To display the range of candidate-level behaviour without selecting only favourable examples, Figure 4 and Table 2 use the highest, median and lowest held-screen Spearman values among the 35 screens containing at least 20 candidates. This case rule was applied after all predictions were frozen and was used only for visualization.

Two diagnostic classifier families quantified residual provenance information but were not used for endpoint prediction. Four structure-availability indicators assigned all 13,605 out-of-fold rows to the correct one of the two sources. Within delivery data, retained non-graph chemistry and context classified the 35 screens containing at least 20 rows under row-level cross-validation ($n = 12,658$; balanced accuracy 0.964). These diagnostics motivated complete group separation while retaining the physical inputs used by the prediction task.

Results

DeepLNP preserves point error while strengthening neural screen ranking

DeepLNP achieved RMSE 1.003 ± 0.003 across the endpoint-balanced five-fold test (Figure 2). The zero and training-mean predictors both gave RMSE 1.003, whereas metadata Extra Trees reached 0.988 ± 0.024 . Thus the neural point head did not improve absolute held-screen error over the null or metadata baseline. Its ranking output was more informative: mean within-screen Spearman was 0.222 ± 0.067 , Hit@10% was 0.178 ± 0.045 and normalized regret was 0.201 ± 0.057 .

Matched baselines separate point and ranking signal

Metadata Extra Trees gave the lowest standalone point error (0.988 ± 0.024) but lower mean screen Spearman (0.197) than DeepLNP. Ionizable-Morgan Extra Trees gave RMSE 1.002 ± 0.036 , Spearman 0.245 ± 0.049 , Hit@10% 0.241 and normalized regret 0.189. The role-wise fingerprint model did not improve on the ionizable-only fingerprint baseline, consistent with the limited helper variation, invariant delivery-side cholesterol and unresolved PEG structure.

The validation-selected hybrid combined the metadata point predictor with the ionizable-Morgan ranking predictor and DeepLNP. It obtained RMSE 0.988 ± 0.021 , Spearman 0.248 ± 0.054 , Hit@10% 0.245 and normalized regret 0.184. Its out-of-fold RMSE difference from metadata Extra Trees was 0.0006 with a screen-bootstrap interval $[-0.0039, 0.0062]$. Relative to DeepLNP, the hybrid changed RMSE by -0.0143 $[-0.0342, 0.0013]$ and screen Spearman by 0.025 $[-0.047, 0.101]$. These intervals support a complementary operating point, not a resolved superiority claim.

Dual-path ablations identify the selected architecture

The graph-only model gave test RMSE 1.000 and mean screen Spearman 0.102 (Table 1; Figure 3). Replacing its ranking score with the direct Morgan pathway raised Spearman to 0.197 but worsened RMSE to 1.047. A direct rank-only configuration reached Spearman 0.245 with RMSE 1.009. The dual residual preserved the graph-derived rank adapter and added a gated Morgan residual; it achieved the best validation screen Spearman (0.209) among the four configurations while retaining validation RMSE 0.999. This validation result selected A115 before the five held-out test folds were summarized.

Held-screen scores prioritize candidates under finite screening budgets

The frozen DeepLNP scores were next evaluated as retrospective candidate-selection decisions rather than only as aggregate prediction metrics. Across 38 held-out screens, selecting the top 5%, 10% and 20% of candidates recovered mean fractions of 0.091, 0.176 and 0.305 of the corresponding measured top sets (Figure 4a). The associated range-normalized regret decreased from 0.278 to 0.203 and 0.100 (Figure 4b). The 95% screen-bootstrap intervals were $[0.051, 0.135]$, $[0.134, 0.220]$ and $[0.255, 0.355]$ for top-set recovery, and $[0.194, 0.372]$, $[0.125, 0.292]$ and $[0.046, 0.169]$ for regret. These values quantify the candidate quality retained when experimental capacity is limited to a fixed fraction of an unseen screen.

Performance varied across biological settings (Figure 4c). Mean screen Spearman was 0.209 for generic-cell screens, 0.162 for liver, 0.183 for lung epithelium, 0.171 for macrophage, 0.312 for muscle and 0.228 for spleen; strata with only two to seven screens had wide bootstrap intervals. Route- and cargo-stratified results, together with every held-screen result, are reported in the Supplementary Material.

At a 10% budget, DeepLNP retained the measured best candidate in both the highest-ranking A549 formulation screen and the median-ranking LM three-component-reaction screen (Table 2; Figure 4d). In the lowest-ranking HBEC-ALI 4CR ketone screen, the three selected candidates did not include FO-8-propyl, and normalized regret was 0.615. The complete candidate table in the Supplementary Material reports the model point output, rank score, formulation ratios and observed response for these decisions.

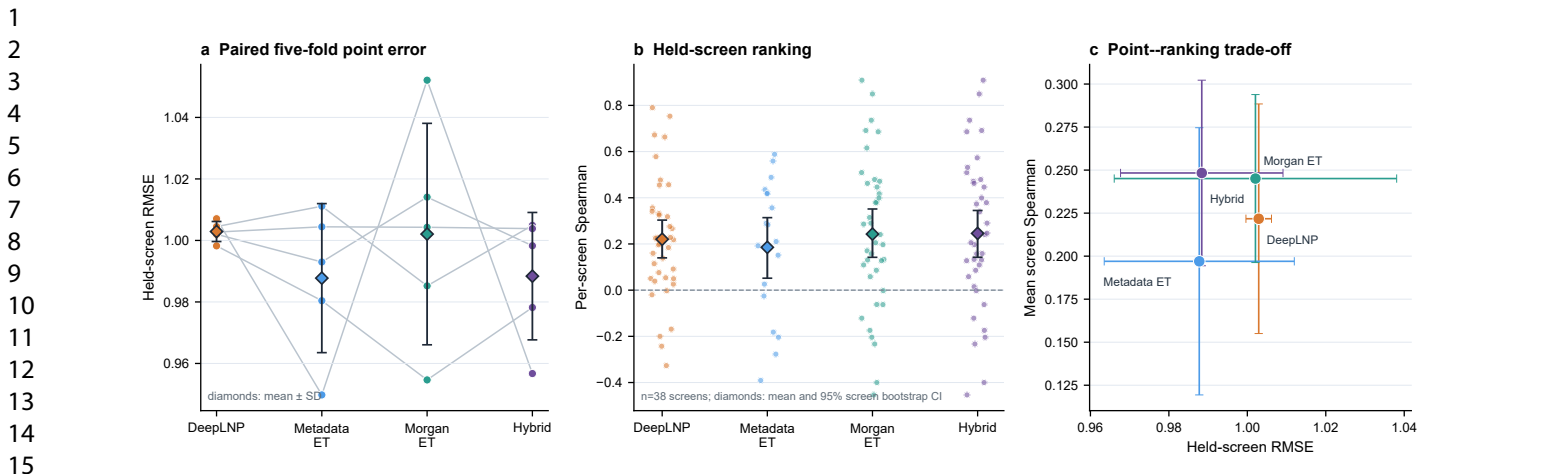


Fig. 2. Held-screen delivery performance under endpoint-balanced five-fold evaluation. **a**, Fold-paired RMSE for DeepLNP, metadata Extra Trees, ionizable-Morgan Extra Trees and the validation-selected hybrid; circles are individual folds and diamonds show mean $\pm$ sample standard deviation. **b**, Defined Spearman correlations for the 38 held-out screens; diamonds and intervals show the equal-screen mean and 95% screen-bootstrap interval. **c**, Fold-mean RMSE versus mean screen Spearman, with bars denoting sample standard deviations across folds.

Alt text: Three panels compare four systems. Metadata Extra Trees and the hybrid have the lowest fold-mean RMSE, while ionizable-Morgan Extra Trees and the hybrid have the highest average held-screen rank correlation; a final scatter displays the point-error and ranking trade-off.

Table 1. Matched five-fold ablation of the DeepLNP ranking pathway. Values are held-test means $\pm$ sample standard deviation. Model selection used validation metrics only.

Configuration	RMSE $\downarrow$	Mean screen ρ $\uparrow$	Hit@10% $\uparrow$	Norm. regret $\downarrow$
Graph pathway only	1.0001 $\pm$ 0.0102	0.1017 $\pm$ 0.1483	0.1183 $\pm$ 0.0471	0.2709 $\pm$ 0.1072
Direct Morgan replaces graph rank	1.0470 $\pm$ 0.0714	0.1968 $\pm$ 0.0706	0.1958 $\pm$ 0.0291	0.1518 $\pm$ 0.0555
Direct Morgan rank only	1.0086 $\pm$ 0.0021	0.2447 $\pm$ 0.0997	0.2422 $\pm$ 0.0358	0.1234 $\pm$ 0.0728
Dual graph-Morgan residual	1.0029 $\pm$ 0.0033	0.2217 $\pm$ 0.0667	0.1778 $\pm$ 0.0447	0.2008 $\pm$ 0.0572

Robustness across split and aggregation choices

In a controlled split-sensitivity run, the learner, feature matrix, hyperparameters and five seeds were held fixed. Row-random tests shared 37.4 ± 0.5 of the 38 screen identities with training and yielded RMSE 0.776 ± 0.024 and mean screen Spearman 0.554 ± 0.032 . Endpoint-balanced group testing enforced zero screen overlap and yielded RMSE 1.035 ± 0.054 and Spearman 0.229 ± 0.097 . The held-screen task is therefore materially harder than interpolation among familiar campaigns.

Development-stage GroupDRO, invariant-risk-minimization and correlation-alignment variants were also tested as shift-robustness ablations [37, 38, 39]. None improved the validation criterion sufficiently to replace the selected dual-path model; their single-split diagnostics are reported in the Supplementary Material and were not used to revise the frozen five-fold result.

Equal-screen aggregation gave Spearman 0.221 for DeepLNP, 0.243 for ionizable-Morgan Extra Trees and 0.246 for the hybrid. Row-size weighting reduced these estimates to 0.195, 0.184 and 0.190, respectively; excluding the largest screen changed each equal-screen estimate by less than 0.002 (Figure 5). Metadata Extra Trees produced defined rank correlations for 18 screens because some held-screen predictions were constant; its defined-screen mean was 0.185. Full weighting and size-stratified estimates are provided in the Supplementary Material.

DeepLNP's nominal 50%, 80%, 90% and 95% Gaussian intervals achieved mean empirical coverages of 0.522, 0.803, 0.900 and 0.949. These descriptive fold averages do not establish conformal coverage or a reliable sample-wise applicability domain under experimental-group shift [40, 41].

Discussion

DeepLNP separates two distinct objectives in LNP screening

DeepLNP was developed around a distinction that is often obscured in LNP prediction: estimating formulation-level responses and prioritizing candidates within an experimental screen are related but not identical objectives. The model therefore combines a role-aware formulation representation with separate point-estimation and screen-ranking outputs, complementing multiobjective LNP engineering that couples prediction with experimental optimization across delivery settings [42]. Across 38 held-out experimental screens, the selected dual graph-Morgan residual increased mean within-screen Spearman correlation from 0.102 for the graph-only neural model to 0.222, while maintaining an RMSE close to one. These results suggest that explicitly allocating model capacity to relative ordering can improve neural candidate ranking without requiring a corresponding improvement in point prediction.

This distinction is particularly relevant because the released delivery labels are standardized within each experimental screen. Consequently, an RMSE near one reflects prediction error on a screen-standardized relative scale rather than absolute delivery efficacy. The ranking output is therefore aligned more directly with the retrospective decision considered here: identifying comparatively strong candidates within a specified experimental context. Increasing the inspection budget from 5% to 20% increased measured top-set recovery and reduced normalized best-candidate regret from 0.278 to 0.100. However, performance remained heterogeneous across screens, including

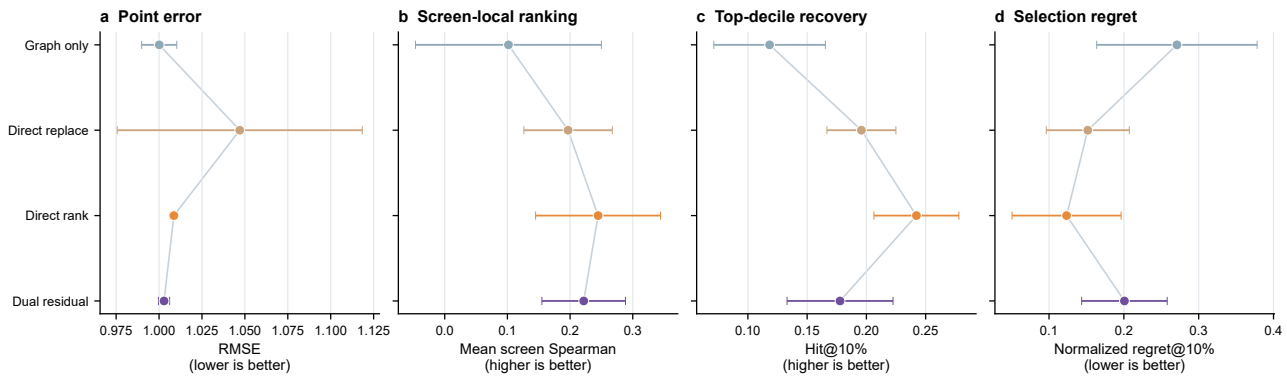


Fig. 3. Five-fold ranking-path ablation. Points and bars show test means $\pm$ sample standard deviation for RMSE, mean screen Spearman, Hit@10% and normalized regret. The direct rank-only pathway maximizes test ranking, while the validation-selected dual residual provides the stronger validation compromise between point and ranking objectives.

Alt text: Four panels compare graph-only, direct replacement, direct rank-only and dual-residual variants. Direct rank-only has the highest test rank correlation; direct replacement has the largest point error; the dual residual remains near the null RMSE with stronger ranking than the graph-only neural model.

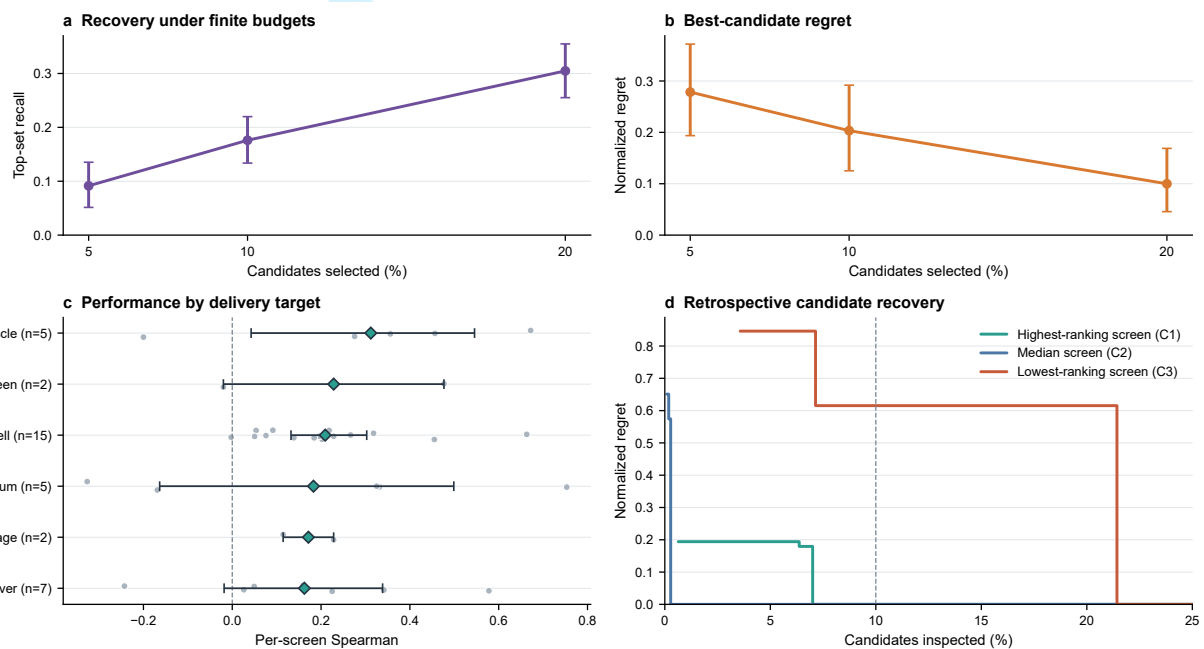


Fig. 4. Retrospective use of frozen DeepLNP scores for held-screen candidate prioritization. **a**, Recovery of the measured top set when 5%, 10% or 20% of each screen is selected. **b**, Range-normalized regret of the best measured candidate retained at the same budgets. Points are equal-screen means across 38 screens and intervals are 95% complete-screen bootstrap intervals. **c**, Screen Spearman grouped by delivery target for strata represented by at least two screens; grey points are screens and diamonds are equal-screen means with 95% bootstrap intervals. **d**, Regret as candidates are inspected in descending DeepLNP rank-score order for the mechanically selected highest, median and lowest ranking cases among screens with at least 20 candidates. The dashed line marks the 10% budget.

Alt text: Four panels show candidate recovery increasing and normalized regret decreasing as the inspection budget grows, variation in screen-level rank correlation across six delivery-target strata, and retrospective regret trajectories for objectively selected high, median and low ranking screens.

a held-out HBEC-ALI example in which candidate ranking failed. Thus, the present evidence supports retrospective prioritization within previously unseen experimental groups, rather than absolute prediction of delivery efficacy or guaranteed recovery of optimal formulations.

Neural and conventional representations capture complementary signals

The comparison with conventional learners provides an important qualification to the neural results. Metadata Extra Trees achieved lower point-prediction error than DeepLNP, whereas ionizable-lipid Morgan Extra Trees achieved stronger mean within-screen ranking. The validation-selected hybrid combined these complementary outputs but did not show a statistically

Table 2. Representative held-screen candidate decisions at a 10% inspection budget. Cases are the highest, median and lowest screen Spearman among screens with at least 20 candidates. Responses are released screen-standardized delivery values, so comparisons are within each row only.

Case	Held-screen setting	n (k)	ρ	Hit@10%	Norm. regret	Best measured selected / best measured available
C1	A549 formulation; lung epithelium; in vitro	157 (16)	0.753	0.188	0.000	Form 22 (2.236) / Form 22 (2.236)
C2	LM three-component reaction; HeLa; in vitro	1080 (108)	0.202	0.324	0.000	LM_3CR_A12Iso52DC18 (4.472) / same candidate (4.472)
C3	4CR ketone; HBEC-ALI; in vitro	28 (3)	-0.327	0.000	0.615	FO-10 (-0.022) / FO-8-propyl (2.517)

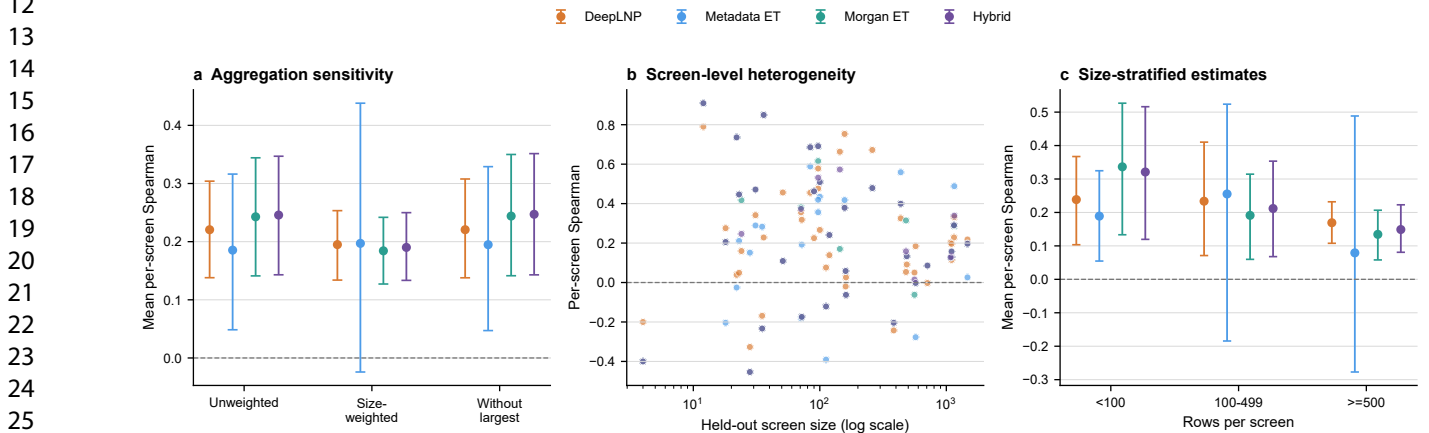


Fig. 5. Sensitivity of held-screen ranking to aggregation and screen size. **a**, Equal-screen, row-size-weighted and largest-screen-excluded estimates. **b**, Per-screen Spearman correlation against held-out screen size. The lowest points arise from one four-candidate screen, where discrete rank changes produce large correlation steps. **c**, Prespecified small, medium and large screen strata; intervals are 95% screen-bootstrap confidence intervals.

Alt text: Three panels compare DeepLNP, metadata Extra Trees, ionizable-Morgan Extra Trees and their hybrid across aggregation rules, individual held-out screen sizes and three screen-size strata. Estimates vary substantially across screens and small-screen intervals are wide.

resolved advantage over the strongest standalone comparator for either task. These findings indicate that model complexity alone does not determine predictive utility in heterogeneous LNP datasets. Different representations appear to exploit different information regimes: formulation and experimental metadata are informative for point estimation, whereas sparse ionizable-lipid substructure features remain highly competitive for relative candidate ordering.

The dual-path architecture can therefore be interpreted as an attempt to preserve both information regimes within a single neural framework. The molecular-graph pathway provides a formulation-level representation in which lipid roles, composition and context can interact, whereas the direct Morgan residual retains a chemically sparse representation that is effective for ranking. Importantly, the direct-rank-only variant achieved the highest test ranking estimate, whereas the dual residual was retained because it provided the better prespecified validation compromise between point and ranking objectives. This result argues against selecting architectures retrospectively on the basis of the strongest test metric and underscores the importance of validation-only model selection when several prediction objectives are considered simultaneously.

More broadly, the competitiveness of tree and fingerprint baselines is consistent with the need for careful benchmarking in molecular and formulation modelling [32]. A complex molecular representation should not be assumed to outperform simpler descriptors when dataset structure, missingness and experimental context themselves carry substantial predictive information.

For LNP modelling, this observation shifts the methodological question from whether deep learning universally outperforms conventional learners to when molecular structure, formulation context and ranking-specific representations contribute complementary information.

Held-screen evaluation exposes experimental-group heterogeneity

A second contribution of this study is the use of complete experimental screens as the unit of held-out evaluation. This design reduces direct sharing of screen identity between training and test sets and more closely reflects the intended setting in which a trained model is applied to a previously unseen screening campaign. The substantial variation in per-screen rank correlation further shows why pooled or row-weighted metrics alone can obscure practically relevant heterogeneity. Equal-screen, size-weighted and largest-screen-excluded analyses produced different ranking estimates, although exclusion of the largest screen had little effect on the equal-screen estimates.

Nevertheless, held-screen evaluation should not be interpreted as evidence of unrestricted biological or chemical transfer. Experimental screens can remain associated with characteristic lipid libraries, targets, assay conditions and patterns of feature availability. Thus, a group-disjoint split prevents direct screen overlap but cannot fully disentangle transferable structure-function relationships from source- or campaign-associated covariates. The current results are therefore most

appropriately interpreted as generalization across the experimental groups represented in the assembled corpus, rather than evidence of universal transfer across targets, lipid chemistries or manufacturing regimes.

Limitations and prospective validation

Several limitations define the scope of the present conclusions. First, structural support across the four canonical lipid roles is highly asymmetric. Delivery-labelled records vary extensively in ionizable-lipid chemistry and include some helper-lipid variation, but delivery-side sterol identity is effectively invariant and the PEG-lipid is not resolved as a molecular graph. The current data therefore support a four-role representation architecture, but do not establish learned sterol- or PEG-specific structure–activity relationships. Second, the auxiliary endpoints are incompletely observed and source-blocked rather than paired with delivery measurements on the same formulations. The masked heads therefore provide a unified interface for heterogeneous endpoint information, but the present experiments do not establish that auxiliary physicochemical or toxicity supervision improves delivery prediction. In particular, the toxicity data comprise only 99 upper-bounded observations from two experimental groups and should be regarded as a demonstration of censored-endpoint handling rather than a validated safety-prediction capability. Third, delivery outcomes are standardized within screen. The model consequently learns relative responses on the released scale and cannot be interpreted as providing calibrated absolute delivery efficacy across assays. Similarly, the retrospective candidate-selection analysis evaluates whether frozen scores would have enriched strong candidates among formulations whose outcomes are already known; it does not constitute prospective experimental validation. The uncertainty estimates should also be interpreted descriptively, because empirical interval coverage does not establish conformal validity or a reliable sample-specific applicability domain under experimental-group shift.

A decisive prospective evaluation would therefore require factorial experiments in which multiple ionizable lipids are crossed with helper lipids, sterols and PEG-lipids at prespecified compositions within the same assay. Shared controls and commensurate raw endpoints would enable absolute calibration and more direct estimation of component interactions. Delivery to target and off-target tissues, dose-dependent viability, inflammatory markers and formulation properties such as particle size, PDI, encapsulation efficiency and stability should ideally be measured on overlapping formulations with biological replication. Model weights, candidate-selection rules and uncertainty thresholds should be frozen before experimental outcomes are revealed. Such a design would provide a stronger test of complete-formulation interactions, biological transfer and prospective candidate prioritization. Spatially informed lipid representations and structure–function databases provide complementary routes for testing physically grounded descriptors when overlapping supervision becomes available [43, 44].

Conclusion

DeepLNP introduces a role-aware dual-path strategy that separates formulation-level prediction from within-screen candidate ranking in LNP modelling. Held-screen evaluation shows that the ranking-specific graph–Morgan residual improves neural

candidate ordering, while conventional metadata and fingerprint baselines remain competitive. The results therefore support objective-specific modelling rather than a single pooled prediction criterion for heterogeneous LNP screens. Because the present evidence is based on screen-standardized relative delivery with incomplete structural and auxiliary-endpoint support, prospective factorial experiments and frozen-model validation are needed to establish complete-formulation interactions, absolute calibration and transferable candidate prioritization.

Key Points

- DeepLNP provides a role-aware formulation representation that integrates lipid roles, molar composition, biological target, assay and process context.
- A dual-path architecture separates formulation-level point prediction from within-screen candidate ranking by combining a shared molecular-graph encoder with a gated ionizable-lipid Morgan residual.
- The ranking-specific residual improved mean within-screen Spearman correlation from 0.102 to 0.222 while maintaining held-screen RMSE at 1.003 ± 0.003 .
- Retrospective candidate prioritization improved with increasing inspection budget, with top-set recovery rising from 0.091 to 0.305 and normalized regret decreasing from 0.278 to 0.100 between 5% and 20% budgets.
- Conventional metadata and fingerprint baselines remained competitive, and current validation is restricted to screen-standardized relative delivery; prospective factorial experiments are required to establish complete-formulation interactions and transferable candidate prioritization.

Author contributions

Z.J.: Conceptualization, methodology, software, formal analysis, visualization and writing—original draft. K.C.: Validation and writing—review and editing. Z.H.: Supervision, project administration, validation and writing—review and editing. All authors reviewed and approved the manuscript.

Supplementary material

Supplementary material is provided as a separate file and contains the complete DeepLNP input schema and active configuration, learning objectives, split construction, fold-level and screen-level results, matched baselines, ranking-path ablations, group-size analyses, calibration, toxicity limitations and reproducibility controls.

Conflict of interest

The authors declare no competing interests.

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Data availability

The analysis-ready corpus integrates LNP_ML data and code distributed through its source repository [45] and LNP Atlas v1 archived under CC BY 4.0 at Zenodo [46]. The designated DeepLNP repository is github.com/YichuanAlex/DeepLNP [47]. The release package prepared for this repository contains the analysis-ready merged table and its SHA-256 checksum, row/group/fold assignments, the 12,692 held-screen out-of-fold predictions for the selected model, aggregate result tables, model and evaluation code, exact random seeds, tested software versions, figure-generation scripts and editable figure sources. Instructions and checksums for reconstructing the merged table from the upstream resources are provided in the repository; the original licences and attribution requirements continue to apply.

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SUPPLEMENTARY MATERIAL

Supplementary material for: DeepLNP: role-aware dual-path learning for lipid nanoparticle delivery efficacy modelling and within-screen ranking

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Abstract

This supplementary material specifies the DeepLNP dual-path architecture and its held-screen evaluation. It documents role-aware molecular and formulation inputs, the graph endpoint pathway, the gated direct Morgan ranking residual, masked multi-endpoint objectives, optimization, matched null and tree baselines, and validation-selected hybrid inference. Extended results include endpoint-balanced five-fold predictions, candidate recovery at fixed experimental budgets, target-, route- and cargo-stratified results, candidate-level model outputs, ranking-path ablations, split and group-size stress tests, interval calibration and the scope of the censored-toxicity interface.

Key words: DeepLNP, lipid nanoparticle, role-aware representation, target conditioning, multi-endpoint learning, screen ranking, group shift

S1. Dataset and prediction tasks

S1.1. Prediction scope

The reported experiments evaluate relative delivery and within-screen ranking; additional overlapping measurements are needed for absolute calibration and prospective optimization. DeepLNP also provides masked heads for particle size, polydispersity index (PDI), zeta potential, encapsulation efficiency and bounded toxicity.

The primary evaluation uses five endpoint-balanced, group-disjoint folds. Earlier group-shuffle development experiments are retained in the project ablation ledger and are not pooled with the final five-fold comparison. A controlled experiment compares row-random and group-disjoint partitions while keeping the learner, feature matrix and hyperparameters fixed. The reporting unit for delivery ranking is an experimental screen; screen-local rank and selection metrics are calculated within each held-out screen before aggregation.

S1.2. Dataset assembly and endpoint support

S1.2.1. Loader-defined analysis population

The analysis population was reconstructed through the project loader from the merged formulation table. A row was retained when it contained a valid primary molecular structure and at least one measured endpoint. The resulting corpus contained 13,605 rows: 12,692 from the high-throughput LNP_ML collection and 913 from the literature-derived LNP Atlas extraction. The delivery endpoint was present in all 12,692 LNP_ML

Table S1. Endpoint counts in the loader-defined corpus.

Endpoint	Available rows
Delivery efficiency	12,692
Particle size	905
Polydispersity index	649
Zeta potential	265
Encapsulation efficiency	570
Numeric or bounded toxicity	99
All records	13,605

The 12,692 delivery rows are distributed across 38 experimental screens.

rows and absent from the LNP Atlas rows. The smaller source contributed the physicochemical and toxicity observations.

Table S1 gives the exact endpoint counts. The delivery rows form 38 experimental screens. Row counts therefore should not be interpreted as independent biological experiments.

S1.2.2. Structural coverage

Each formulation may contain molecular structures for an ionizable lipid, helper lipid, sterol and PEG-lipid. Structure validity was established by molecular parsing after deterministic resolution of verified lipid names. Product labels that did not determine a unique chemical structure were left missing. In

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Table S2. Feature support conditional on endpoint. Counts are followed by one-decimal percentages for partial support; complete and absent support are shown as “all” and “none”.

Feature block	Delivery	Size	PDI	Zeta	Encapsulation	Toxicity
Ionizable structure	12,692 (all)	900 (99.4%)	644 (99.2%)	263 (99.2%)	565 (99.1%)	99 (all)
Helper structure	12,620 (99.4%)	870 (96.1%)	635 (97.8%)	249 (94.0%)	559 (98.1%)	99 (all)
Sterol structure	12,093 (95.3%)	890 (98.3%)	644 (99.2%)	264 (99.6%)	565 (99.1%)	99 (all)
PEG-lipid structure	0 (none)	771 (85.2%)	516 (79.5%)	232 (87.5%)	534 (93.7%)	99 (all)
Any composition/process field	12,692 (all)	905 (all)	649 (all)	265 (all)	570 (all)	99 (all)
Biological-target evidence	12,692 (all)	770 (85.1%)	607 (93.5%)	240 (90.6%)	477 (83.7%)	99 (all)

Table S3. Delivery-side component support at four distinct levels. Activity denotes a positive reported molar fraction; identity and graph columns count rows rather than unique molecules.

Role	Active rows	Row-level ID	Graph resolved	Active + graph	Unique graphs
Ionizable lipid	12,692	12,692	12,692	12,692	9,194
Helper lipid	12,133	12,620	12,620	12,133	4
Sterol	12,093	12,093	12,093	12,093	1
PEG-lipid	12,021	0	0	0	0

The invariant cholesterol graph supplies sterol amount and identity information but no sterol chemical variation. “Row-level ID” counts identities present in the merged model table, not identities inferred from a subset of source-level protocols. PEG activity is known from the reported molar fraction; zero resolved graphs therefore means unavailable discrete structure, not absence of PEG from the formulation.

particular, the loader did not infer a PEG chain length from a nominal-mass product name.

The corpus contained 732 rows with resolved structures for all four roles. None of these rows had a delivery label. Among delivery rows, ionizable-lipid graphs were resolved in all 12,692 cases and helper-lipid graphs in 12,620 cases. The source protocol specifies cholesterol for 12,093 positive-sterol formulations, yielding one delivery-side sterol graph. The merged LNP_ML rows report PEG molar fraction but contain neither a PEG-identity field nor a PEG SMILES. Formulation comments identify C14-P2K in 84 retained Lee rows and generic PEG2000 in 24 retained Zhou rows; these product-average labels still do not define unique molecular graphs and do not establish the PEG identity of the remaining screens. We therefore retained PEG amount and activity while leaving the delivery-side PEG graph unresolved. Table S2 gives conditional counts and coverage. The formulation/process entry means that at least one structured field was present in the block; it does not mean that every process quantity was observed. Biological-target evidence denotes a raw delivery target or a non-empty biodistribution or gene-expression result; an encoded missing category is not counted as support.

S1.2.3. Endpoint overlap

Delivery does not co-occur row-wise with any auxiliary endpoint (Table S4 and Figure S1). The auxiliary physicochemical endpoints overlap extensively with one another because they arise from the same literature-derived formulations. Consequently, masked multi-endpoint training does not supply paired physicochemical supervision for the delivery rows. It provides a common model interface across source-blocked tasks.

S1.2.4. Delivery-label semantics

For delivery observation n in upstream screen $g(n)$, the released target is standardized within that screen:

$$y_n = \frac{a_n - \bar{a}_{g(n)}}{s_{g(n)}}, \quad s_g^2 = \frac{1}{|g|} \sum_{m \in g} (a_m - \bar{a}_g)^2. \quad (S1)$$

The 12,692 released delivery labels have mean -0.0055 , standard deviation 1.0026 and range $[-3.157, 7.224]$. Accordingly, RMSE is measured in upstream within-screen standard deviations, and validation concerns the released relative endpoint and within-screen ranking rather than absolute activity.

S2. DeepLNP architecture

S2.1. Input schema and missingness representation

S2.1.1. Structured input blocks

Table S5 documents the model-facing schema. The four lipid roles are kept distinct even though graph-encoder weights are shared. Missing structures use a one-node dummy graph and zero descriptors, while a second mask records positive molar fraction. This pattern preserves implicit missingness when the explicit validity embedding is disabled.

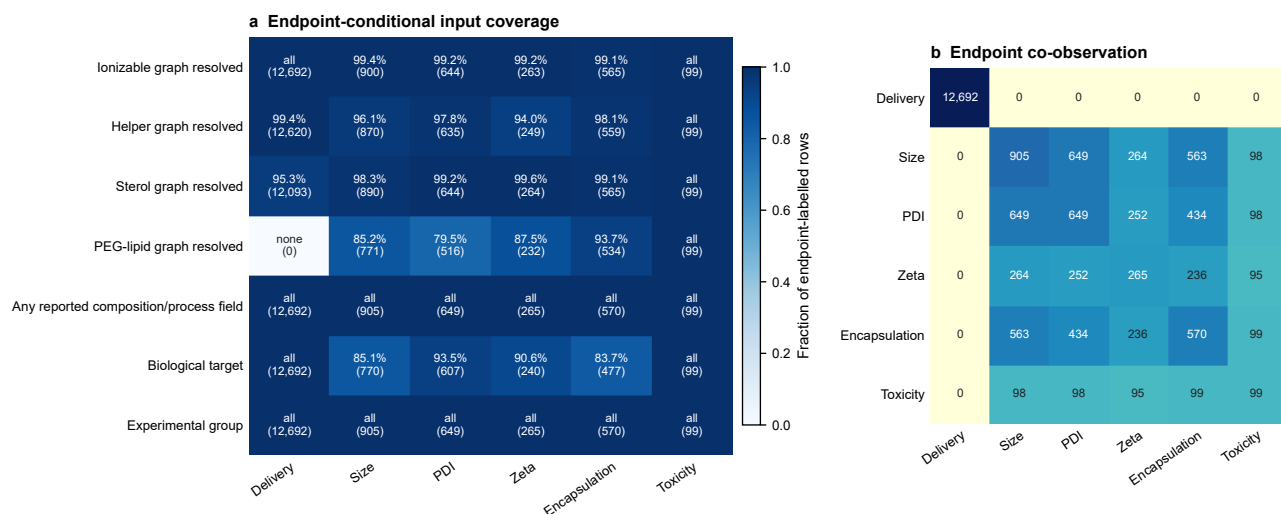
S2.1.2. Process missingness

Every one of the 14 process values is paired with an observed/missing indicator. If a value is unavailable, its scaled numeric entry is set to zero while its indicator is zero. A reported physical zero retains a zero numeric entry but has indicator one. This representation distinguishes an unreported condition from, for example, a formulation with 0% helper lipid. Missing molecular structures use the dummy-graph and zero-descriptor pattern, but their validity bits are not concatenated as predictive features in the selected model. A positive reported molar fraction, rather than structure validity, controls whether a role token participates in attention. The implicit missingness pattern remains learnable even without the explicit embedding. Scaling constants are fixed by field type; endpoint normalization is fitted only on training rows.

Explicit source identity is excluded, although molecular libraries, targets, routes and missingness patterns retain indirect provenance information; Section S4.4 quantifies this dependence.

Table S4. Pairwise counts of rows with both endpoints observed.

Endpoint	Delivery	Size	PDI	Zeta	Encapsulation	Toxicity
Delivery	12,692	0	0	0	0	0
Size	0	905	649	264	563	98
PDI	0	649	649	252	434	98
Zeta	0	264	252	265	236	95
Encapsulation	0	563	434	236	570	99
Toxicity	0	98	98	95	99	99



Composition/process support is a logical OR across reported fields, not complete process coverage.

Fig. S1. Feature support conditional on endpoint and pairwise endpoint overlap. Delivery has no overlap with complete four-role structures or auxiliary endpoints.

Alt text: The left heat map shows feature availability for six endpoints. The right heat map shows pairwise endpoint overlap. Delivery occupies a separate source block from the physicochemical and toxicity measurements.

S2.2. Role-aware component interaction

Let $i \in \{1, 2, 3, 4\}$ denote the ionizable lipid, helper phospholipid, sterol and PEG-lipid roles. A shared two-layer graph encoder G_θ transforms each valid molecular graph into a 64-dimensional vector. Role identity is introduced after graph encoding so that the chemistry encoder is shared while the downstream model can distinguish functional positions. The initial token is

$$\mathbf{h}_{ni}^{(0)} = G_\theta(S_{ni}) + \mathbf{e}_i + \phi(r_{ni}) + D(\mathbf{d}_{ni}), \quad (\text{S2})$$

where $\mathbf{e}_i$ is a learned role embedding, ϕ is a two-layer molar-fraction encoder and D projects the six deterministic chemical descriptors. The selected graph pathway disables learned structure-missingness embeddings and pairwise attention bias. A separate ionizable-lipid Morgan branch is enabled only for the ranking residual. Two four-head pre-normalized self-attention blocks update the four active role tokens.

The target-conditioned representation uses assay and biological context as a query over the role tokens. Cargo, model system and route are encoded separately from the 11-class biological-target context. The target query is added only after an unconditional role-token pooling operation. This yields both a conditioned molecular representation for endpoint prediction and an unconditional representation for the auxiliary target classifier. The latter prevents target-class supervision from reading the target one-hot vector directly.

The global formulation branch encodes the 14 process and composition values together with their 14 observation indicators. Molar fractions therefore enter through two deliberate paths: role-token composition controls molecular interactions, while the global branch retains composition alongside N/P ratio, mixing and dose. The molecular and global branches are projected to 64 dimensions and integrated by a three-expert gated fusion module. Endpoint-specific mean and log-variance heads operate on the fused representation. Delivery ranking also receives a direct ionizable-fingerprint residual described below.

S2.2.1. Spatial feature scope

The optional 12-dimensional role descriptor contains six deterministic chemical quantities and six single-conformer shape quantities. The conformer is generated with ETKDGv3 using a fixed random seed. A failed or unavailable conformer produces zeros, which are interpreted together with the structure-validity mask. The reported five-fold experiments disabled conformer generation, so the six shape channels are zero and provide no three-dimensional molecular evidence. When enabled, these descriptors represent one isolated computational conformer rather than lipid protonation ensembles, intermolecular assembly, solvent, cargo packing or dynamic LNP morphology.

Table S5. DeepLNP input and supervision schema. “Primary” refers to the endpoint-balanced neural evaluation; comparator-only and optional branches are identified explicitly.

Block	Dimension count	or	Primary use	Definition
Molecular roles	Four		Yes	Ionizable lipid, helper phospholipid, sterol and PEG-lipid occupy separate role slots.
Molecular graph	39 atom; 4 bond features		Yes	The atom vector has 21 populated channels for identity, atomic number, degree, charge, hydrogen count, aromaticity, ring membership and hybridization, followed by 18 fixed zero-padding channels; bonds use single, double, triple and aromatic indicators.
Morgan representation	2,048		Ranking path	Radius-two chiral count fingerprint of the ionizable lipid, transformed by $\log(1+x)$; used by the direct ranking residual and fingerprint baselines.
Chemical descriptors	Six per role		Yes	Molecular weight, logP, topological polar surface area, fraction sp^3 , rotatable-bond count and formal charge.
Conformer descriptors	Six per role		No	Radius of gyration, asphericity, eccentricity, inertial shape factor, NPR1 and NPR2; disabled in the primary evaluation.
Role masks	Four plus four		Partial	Structure validity is retained for bookkeeping and graph construction but is not added as a predictive token feature; positive molar fraction controls token activity.
Molar composition	Four		Yes, two paths	Fractions enter both the role-token encoder and the global formulation branch.
Formulation/process	14 values plus 14 indicators		Yes	Four molar ratios, four mass ratios, ionizable-lipid:mRNA weight ratio, N/P ratio, aqueous:organic ratio, total flow rate, microfluidic mixing and dose.
Assay context	22 non-source one-hot columns		Yes	Cargo (5), model system (11) and route (6); the four-position source block is fixed to zero.
Biological-target context	11		Yes	One-hot closed vocabulary: generic cell, macrophage, liver, muscle, lung, lung epithelium, spleen, tumour, brain, other and missing. Literature-derived rows use explicit biodistribution or gene-expression evidence when available; 138 rows without target evidence retain the missing category.
Endpoint supervision	Six values plus masks		Yes	Delivery, size, zeta potential, PDI, encapsulation and toxicity are accompanied by row-level observation masks.
Toxicity censoring	One signed value		Interface	The interface supports upper, lower and approximately exact observations; all 99 retained toxicity records are upper bounds.

S3. Training objectives and optimization

S3.1. Partially observed endpoint likelihood

For endpoint q , DeepLNP predicts mean μ_{nq} and log variance ℓ_{nq} . If Ω_q is the set of rows with an observed value, the endpoint contribution is

$$\mathcal{L}_q = \frac{1}{|\Omega_q|} \sum_{n \in \Omega_q} \left[\frac{(y_{nq} - \mu_{nq})^2}{2 \exp(\ell_{nq})} + \frac{\ell_{nq}}{2} \right]. \quad (\text{S3})$$

The endpoint weights are 1.0 for delivery, 0.5 for size, 0.3 for zeta potential, 0.3 for PDI, 0.4 for encapsulation and 0.7 for toxicity. A row with no label for endpoint q contributes nothing to $\mathcal{L}_q$.

All 99 retained toxicity observations are upper bounds on risk. For bound u_n , the censored contribution is

$$\mathcal{L}_n^{\text{upper}} = -\log \Phi \left(\frac{u_n - \mu_n}{\exp(\ell_n/2)} \right), \quad (\text{S4})$$

where Φ is the standard normal cumulative distribution function [33]. The software interface also supports lower bounds and approximately exact observations, but neither occurs in the retained toxicity subset.

S3.2. Screen-aware ranking objective

The graph pathway produces a gated ranking residual over the point mean,

$$r_n^{\text{graph}} = \text{stopgrad}(\mu_n) + \sigma(\gamma) a_\psi(\mathbf{z}_n), \quad (\text{S5})$$

where $\mathbf{z}_n$ is the fused representation, σ is the sigmoid function and γ is a learned scalar gate. The delivery mean is detached in this branch, whereas ranking gradients update the shared representation through $a_\psi(\mathbf{z}_n)$. The direct ranking input $\mathbf{u}_n$ concatenates the 2,048-dimensional log-count ionizable-lipid fingerprint, 12 descriptor positions, 28 formulation/process positions, 26 assay-context positions, 11 target positions and eight role masks. A two-layer perceptron and a second learned gate produce

$$r_n = r_n^{\text{graph}} + \sigma(\eta) b_\omega(\mathbf{u}_n). \quad (\text{S6})$$

The direct branch has no connection to the endpoint mean. For comparable pairs $\mathcal{P}_B$ drawn from same-screen subsets inside minibatch B , the primary pair contribution is logistic:

$$\mathcal{L}_{\text{pair}} = \frac{1}{|\mathcal{P}_B|} \sum_{(i,j) \in \mathcal{P}_B} \text{softplus} [-\text{sgn}(y_i - y_j)(r_i - r_j)]. \quad (\text{S7})$$

Pairs whose difference in the train-standardized delivery target is at most 0.1 are excluded. For each within-minibatch screen

Table S6. DeepLNP configuration used in the endpoint-balanced five-fold evaluation.

Block	Setting	Selection status
Graph encoder	Two layers, width 64	Enabled; shared across four roles
Component interaction	Two blocks, four heads	Enabled; pairwise attention bias disabled
Role inputs	Role and linear ratio encodings	Enabled; predictive structure-validity embedding disabled
Chemical input	Molecular graph plus six 2D descriptors; ionizable Morgan rank input	Graph pathway predicts endpoint means; Morgan pathway contributes only to delivery rank
Context fusion	Width 64, three experts, dropout 0.1	Enabled; explicit source identity disabled
Prediction heads	Six mean/log-variance pairs	Masked by endpoint availability
Ranking branch	Graph residual plus gated direct Morgan residual; margin 0.1	Pair weight 1.0; centred weight 0.5
Auxiliary target head	11 classes	Cross-entropy weight 0.2 on non-missing classes
Optimization	AdamW, learning rate 5×10^{-4} , weight decay 10^{-4}	Mixed precision; gradient-norm limit 1.0
Training budget	Maximum six epochs; patience two	Checkpoint selected on validation groups only
Trainable parameters	529,415	Selected dual-residual configuration

subset with at least three labelled candidates, the centred term is one minus the cosine similarity between mean-centred predicted and measured vectors. The group-pair sampler selects four delivery screens and four rows per screen, sampling rows with replacement when a screen contains fewer than four eligible rows. These 16 pair-focused positions occupy half of a 32-row batch; the remaining 16 positions are filled with replacement from rows carrying at least one auxiliary endpoint label. The centred-term weight is 0.5 and the pair-term weight is 1.0. ListMLE was evaluated during development and excluded from the selected five-fold model.

An auxiliary 11-class target classifier receives the unconditional molecular representation. Its cross-entropy weight is 0.2 and its loss is applied only for non-missing target classes. The complete selected objective is the weighted sum of available endpoint likelihoods, delivery pair and centred losses, and target-class cross-entropy. The efficacy-only ablation changes only the enabled endpoint set and sets the target auxiliary weight to zero.

S3.3. Frozen primary configuration

The validation checkpoint minimizes delivery RMSE minus 0.10 times pooled Spearman, 0.10 times mean screen Spearman, 0.05 times pair accuracy and 0.05 times pooled/global NDCG. The primary fivefold runs reproduce this recorded scalarization exactly. This validation-only scalarization serves checkpoint selection; its global NDCG term is distinct from the screen-equal NDCG reported in the result tables.

S4. Baselines and group-aware evaluation

S4.1. Comparators and hybrid inference

The matched baseline panel uses a common five-fold split and validation-only leaf-size selection. The zero and training-mean predictors establish the scale of the screen-standardized endpoint. Metadata Extra Trees receives molar fractions, formulation/process values and observation indicators, assay and target context, and role-validity/activity masks. Ionizable-Morgan Extra Trees adds the 2,048-dimensional log-count fingerprint of the ionizable lipid. Role-wise-Morgan Extra Trees adds all

four role fingerprints where graphs are resolved. Each tree contains 300 estimators and selects minimum leaf size from {2, 5, 10} on validation groups. Explicit source, publication, screen and split identifiers are excluded.

The hybrid uses metadata Extra Trees as the tree point predictor and ionizable-Morgan Extra Trees as the tree ranker. Point and ranking weights are chosen independently from {0, 0.1, ..., 1} on validation groups. Point weights minimize RMSE; ranking weights maximize mean screen Spearman after within-batch score standardization.

S4.2. Group-separated partitions and leakage controls

S4.2.1. Group definition

The primary group key uses the most specific available experimental identifier. Delivery rows use the upstream normalization-screen key, which combines experiment/library, target, model system and route. Literature rows use a DOI or paper title when no screen identifier is available, and fall back to source only if no finer identifier exists. Group assignment is performed on the complete 13,605-row corpus before endpoint filtering.

S4.2.2. Endpoint-balanced five-fold construction

The fold algorithm first randomizes groups with a fixed seed, then sorts delivery-bearing groups by delivery-row count. Each complete group is assigned to the fold with the smallest current delivery count subject to a balanced number of delivery screens. Non-delivery publication groups are assigned afterward to balance total corpus rows. Only group identity, group size and endpoint availability are used. Endpoint values are not used during fold construction.

The corpus size is divisible by five, and each fold contains 2,721 total rows. Delivery support ranges from 2,532 to 2,548 rows, with seven or eight delivery screens per fold (Table S7). Fold k is used as test and fold $k + 1$ modulo five as validation; the remaining three folds are training data. Thus, every delivery screen is used once for test and once for validation across the five runs.

Table S7. Test-fold support in the endpoint-balanced five-fold evaluation.

Fold	Corpus rows	Delivery rows	Delivery screens
1	2,721	2,548	8
2	2,721	2,532	7
3	2,721	2,537	7
4	2,721	2,541	8
5	2,721	2,534	8

S4.2.3. Preprocessing and selection controls

Endpoint mean and standard deviation are fitted on training rows and applied to validation and test rows. Missing-value indicators are constructed before model fitting and are not recomputed from held-out labels. DeepLNP checkpoint selection and hybrid weights use validation groups only. Classical and neural models receive identical train, validation and test row assignments. Test labels are opened only for final metric calculation.

S4.3. Metrics and aggregation

For predictions $\hat{y}_i$ and released targets y_i , point error is summarized by

$$\text{RMSE} = \left(\frac{1}{N} \sum_{i=1}^N (\hat{y}_i - y_i)^2 \right)^{1/2}. \quad (\text{S8})$$

Pooled Spearman correlation is calculated across all rows in a held-out fold. It is reported separately from the mean screen Spearman

$$\bar{\rho}_{\text{screen}} = \frac{1}{G} \sum_{g=1}^G \rho_S(\hat{\mathbf{y}}_g, \mathbf{y}_g), \quad (\text{S9})$$

where every screen has equal weight, independent of its row count.

Hit@10% measures recovery of the measured top decile by the predicted top decile. If $T_{g,0.1}$ and $\hat{T}_{g,0.1}$ are the observed and predicted top-decile sets,

$$\text{Hit@10\%}_g = \frac{|T_{g,0.1} \cap \hat{T}_{g,0.1}|}{|\hat{T}_{g,0.1}|}. \quad (\text{S10})$$

Selection regret compares the best observed response among selected candidates with the best response available in the screen:

$$R_{g,0.1}^{\text{norm}} = \frac{\max_{i \in g} y_i - \max_{i \in \hat{T}_{g,0.1}} y_i}{\max_{i \in g} y_i - \min_{i \in g} y_i}. \quad (\text{S11})$$

Lower normalized regret is better. The NDCG reported in the result tables is calculated within each screen and averaged with equal screen weight. Pair accuracy constructs comparable pairs only within a screen using a margin of 0.1 in released delivery-target units, then pools correct and total pair counts across screens; it is therefore weighted by the number of comparable pairs. Pooled Spearman and pooled/global NDCG are secondary diagnostics, and the latter also enters the recorded validation checkpoint scalarization described above.

Primary summary values are fold means with sample standard deviations. Screen-bootstrap confidence intervals resample complete held-out screens with replacement. The group-size analyses additionally compare equal-screen weighting, row-count weighting, exclusion of the largest screen and prespecified size strata.

Table S8. Diagnostic source- and screen-classification results.

Diagnostic input	Diagnostic outcome
Two-source classification	
Four structure-availability indicators	13,605/13,605 correct
Thirty-five eligible screens (≥ 20 rows; $n = 12,658$)	
Process-observation indicators	Balanced accuracy 0.057
Process values	Balanced accuracy 0.546
Context plus target	Balanced accuracy 0.515
Role masks plus molar ratios	Balanced accuracy 0.418
Ionizable-lipid fingerprint	Balanced accuracy 0.676
All retained non-graph inputs	Balanced accuracy 0.964

The source count is the number of correct out-of-fold assignments. Screen results use row-level cross-validation and include only screens with at least 20 rows. Explicit dataset, paper, screen and split identifiers were excluded from the predictive schema. High diagnostic accuracy from retained physical inputs motivates group-disjoint evaluation; attribution to individual features was not evaluated.

S4.4. Provenance and screen identifiability

Diagnostic classifiers were fitted to measure whether data support and retained inputs encode source or screen identity. They are not downstream performance models. Table S8 reports complete-corpus out-of-fold separation for the two-source audit and five-fold row-level balanced accuracy for the screen audit.

The deterministic two-source separation follows from source-blocked structural support. The row-level within-delivery diagnostic shows that chemical libraries and assay contexts are distinguishable across the 35 sufficiently supported screens. Evaluation therefore uses complete experimental-group separation and excludes explicit provenance fields.

S5. Extended predictive results

S5.1. Endpoint-balanced five-fold evaluation

S5.1.1. Aggregate results

Tables S9 and S10 extend the main-text result with pooled Spearman, screen-local NDCG and pair accuracy. All four systems use identical held-screen folds; all hyperparameters and hybrid weights are selected from validation groups.

Metadata Extra Trees gave the lowest fold-mean RMSE. The hybrid and ionizable-Morgan tree gave the strongest screen-local selection estimates, while DeepLNP remained close in ranking and improved substantially over its graph-only ablation.

S5.1.2. Fold-level results

Table S11 reports the two primary task metrics for every model and fold. Full predictions and secondary metrics are retained in the machine-readable result package.

S5.2. Retrospective screening utility and candidate-level outputs

The frozen out-of-fold DeepLNP rank scores were converted into explicit screening decisions by retaining the top $k = \lceil qn_g \rceil$ candidates from every held-out screen at budgets $q \in \{0.05, 0.10, 0.20\}$. Across the 38 screens, mean top-set recovery was 0.091, 0.176 and 0.305, while mean range-normalized regret was 0.278, 0.203 and 0.100. These are equal-screen estimates; their complete-screen bootstrap intervals appear in the main-text application figure.

Table S9. Aggregate point and rank-correlation test metrics for the endpoint-balanced five-fold evaluation. Values are fold mean $\pm$ sample standard deviation.

Model	RMSE	Pooled ρ	Screen ρ	Screen-local NDCG
DeepLNP	1.0029 $\pm$ 0.0033	0.1170 $\pm$ 0.0994	0.2217 $\pm$ 0.0667	0.8218 $\pm$ 0.0386
Metadata Extra Trees	0.9877 $\pm$ 0.0242	0.0844 $\pm$ 0.1448	0.1970 $\pm$ 0.0776	0.7972 $\pm$ 0.0363
Ionizable-Morgan Extra Trees	1.0021 $\pm$ 0.0360	0.1707 $\pm$ 0.0729	0.2451 $\pm$ 0.0488	0.8274 $\pm$ 0.0432
Validation-selected hybrid	0.9884 $\pm$ 0.0207	0.1253 $\pm$ 0.1278	0.2483 $\pm$ 0.0539	0.8290 $\pm$ 0.0441

Table S10. Aggregate screen-local decision metrics for the same evaluation. Values are fold mean $\pm$ sample standard deviation.

Model	Pair accuracy	Hit@10%	Normalized regret@10%
DeepLNP	0.5821 $\pm$ 0.0315	0.1778 $\pm$ 0.0447	0.2008 $\pm$ 0.0572
Metadata Extra Trees	0.2166 $\pm$ 0.2388	0.1621 $\pm$ 0.0948	0.2543 $\pm$ 0.0967
Ionizable-Morgan Extra Trees	0.5777 $\pm$ 0.0303	0.2413 $\pm$ 0.0515	0.1889 $\pm$ 0.0384
Validation-selected hybrid	0.5805 $\pm$ 0.0319	0.2449 $\pm$ 0.0469	0.1839 $\pm$ 0.0379

All three metrics restrict comparisons or selections to candidates from the same screen. Hit@10% and normalized regret are averaged with equal screen weight; pair accuracy pools correct and total comparable-pair counts. Constant metadata-tree predictions in some screens yield tied pair decisions and undefined Spearman correlations. Boldface marks the best mean without implying significance.

Table S11. Fold-level held-screen RMSE and mean screen Spearman.

Model	Fold 1		Fold 2		Fold 3		Fold 4		Fold 5	
	RMSE	ρ	RMSE	ρ	RMSE	ρ	RMSE	ρ	RMSE	ρ
DeepLNP	1.0045	0.2711	1.0071	0.2987	0.9982	0.1906	1.0027	0.2185	1.0020	0.1298
Metadata Extra Trees	1.0111	0.2011	0.9498	0.2210	0.9804	0.0908	1.0044	0.1683	0.9930	0.3038
Ionizable-Morgan Extra Trees	0.9852	0.2301	1.0521	0.3033	0.9547	0.2750	1.0043	0.1742	1.0141	0.2429
Validation-selected hybrid	1.0049	0.2301	0.9567	0.3195	0.9782	0.2750	1.0038	0.1742	0.9983	0.2429

Test-fold delivery support was 2,548, 2,532, 2,537, 2,541 and 2,534 rows, respectively. The folds contained 8, 7, 7, 8 and 8 held-out screens.

Table S12. DeepLNP candidate-prioritization results across biological and experimental contexts. Means give every held-out screen equal weight. Spearman intervals are 95% complete-screen bootstrap intervals; no interval is reported for a one-screen stratum.

Context	Stratum	Screens	Rows	Mean screen ρ [95% CI]	Hit@10%	Norm. regret
<i>Delivery target</i>						
	Dendritic cell	1	12	0.790 [-]	0.000	0.114
	Generic cell	15	8239	0.209 [0.132, 0.303]	0.196	0.191
	Liver	7	810	0.162 [-0.018, 0.339]	0.194	0.181
	Lung	1	22	0.039 [-]	0.333	0.382
	Lung epithelium	5	1812	0.183 [-0.163, 0.499]	0.146	0.200
	Macrophage	2	1136	0.171 [0.115, 0.228]	0.211	0.123
	Muscle	5	404	0.312 [0.042, 0.546]	0.128	0.308
	Spleen	2	257	0.228 [-0.020, 0.477]	0.131	0.157
<i>Administration route</i>						
	In vitro	22	11164	0.243 [0.145, 0.348]	0.175	0.177
	Intramuscular	5	404	0.312 [0.026, 0.543]	0.128	0.308
	Intratracheal	1	35	-0.169 [-]	0.250	0.319
	Intravenous	10	1089	0.163 [0.018, 0.313]	0.195	0.197
<i>Cargo</i>						
	mRNA	31	9287	0.228 [0.131, 0.329]	0.178	0.209
	pDNA	1	112	0.076 [-]	0.083	0.438
	siRNA	6	3293	0.207 [0.148, 0.260]	0.180	0.134

Table S12 separates the frozen model result by delivery target, administration route and cargo. These strata describe the contexts represented by the held screens and are not independent transfer experiments because chemistry libraries and assay conditions vary between screens. Figure S2 and Table S13 retain every screen, including small or adverse cases.

The representative cases in the main text were selected after prediction as the maximum, median and minimum screen

Spearman among the 35 screens with at least 20 candidates; this rule did not enter model fitting or checkpoint selection. Table S14 reports the point output, rank score, formulation ratios and observed response, with the separate rank score determining candidate order.

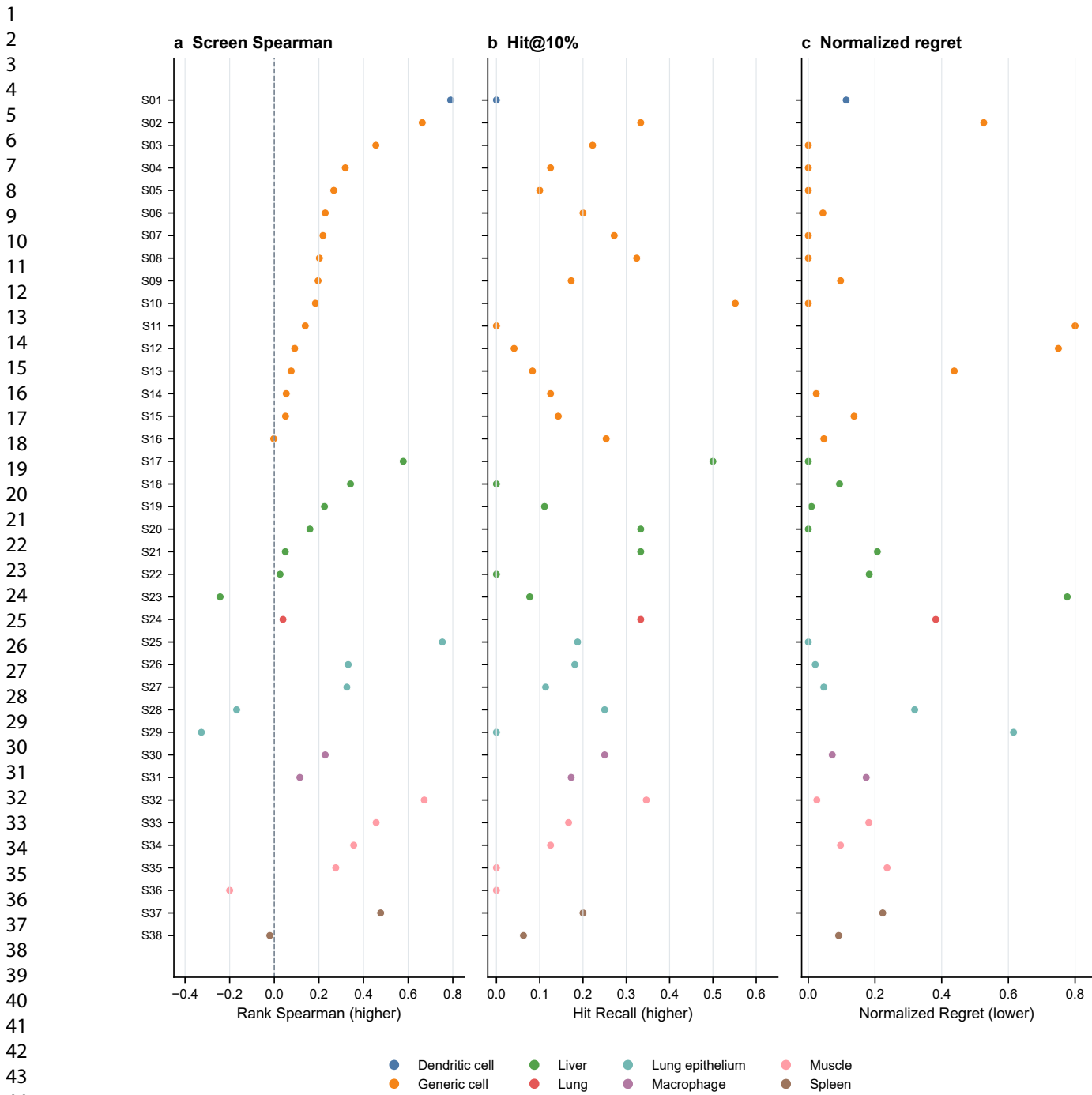


Fig. S2. Complete held-screen distribution of frozen DeepLNP decision metrics. Each row is one of the 38 screens listed in Table S13; colour denotes delivery target. **a.** Screen-local Spearman correlation. **b.** Recovery of the measured top decile by the predicted top decile. **c.** Range-normalized regret of the best candidate in the predicted top decile.

Alt text: Three aligned dot plots display Spearman correlation, top-decile recovery and normalized regret for all 38 held-out screens. Results range from strong ranking with zero regret to negative ranking and high regret, and colours identify eight delivery-target strata.

Table S13. Complete screen-level DeepLNP candidate-prioritization results. Screen codes correspond to Figure S2; metrics are computed from the frozen out-of-fold test prediction for each screen.

Code	Experimental library	Target; model; route	Cargo	<i>n</i>	ρ	Hit@10%	Norm. regret
S01	LM 3CR	Dendritic cell; BMDM; in vitro	mRNA	12	0.790	0.000	0.114
S02	Enzyme-catalysed library	Generic cell; HEK293T; in vitro	mRNA	144	0.663	0.333	0.526
S03	Unsaturated dendrimer	Generic cell; IGROV1; in vitro	mRNA	84	0.455	0.222	0.000
S04	Zwitterionic lipid	Generic cell; HeLa; in vitro	siRNA	72	0.318	0.125	0.000
S05	Disulfide lipid	Generic cell; IGROV1; in vitro	mRNA	100	0.267	0.100	0.000
S06	Whitehead lipidoid	Generic cell; model not reported; in vitro	siRNA	1150	0.228	0.200	0.044
S07	Dendrimer lipid	Generic cell; HeLa; in vitro	siRNA	1470	0.218	0.272	0.000
S08	LM 3CR	Generic cell; HeLa; in vitro	mRNA	1080	0.202	0.324	0.000
S09	AGILE 3CR	Generic cell; HeLa; in vitro	mRNA	1100	0.196	0.173	0.097
S10	iPhos	Generic cell; IGROV1; in vitro	mRNA	572	0.184	0.552	0.000
S11	Aza-Michael	Generic cell; IGROV1; in vitro	mRNA	119	0.139	0.000	0.800
S12	Michael addition	Generic cell; HeLa; in vitro	siRNA	487	0.091	0.041	0.750
S13	Thiol-yne	Generic cell; HEK293T; in vitro	pDNA	112	0.076	0.083	0.438
S14	STING 3CR	Generic cell; HeLa; in vitro	mRNA	480	0.054	0.125	0.024
S15	Aldehyde 4CR	Generic cell; HeLa; in vitro	mRNA	560	0.050	0.143	0.137
S16	Carbonate lipid	Generic cell; model not reported; in vitro	mRNA	709	-0.002	0.254	0.047
S17	Branched ester	Liver; mouse; intravenous	mRNA	97	0.578	0.500	0.000
S18	Disulfide lipid	Liver; mouse; intravenous	mRNA	31	0.342	0.000	0.094
S19	Whitehead lipidoid	Liver; model not reported; intravenous	siRNA	90	0.225	0.111	0.009
S20	Dendrimer lipid	Liver; mouse; intravenous	siRNA	24	0.160	0.333	0.000
S21	Three-form branched ester	Liver; mouse; intravenous	mRNA	23	0.049	0.333	0.207
S22	He aldehyde 4CR	Liver; mouse; intravenous	mRNA	161	0.026	0.000	0.183
S23	Expanded branched ester	Liver; mouse; intravenous	mRNA	384	-0.243	0.077	0.777
S24	Branched ester	Lung; mouse; intravenous	mRNA	22	0.039	0.333	0.382
S25	A549 formulation	Lung epithelium; A549; in vitro	mRNA	157	0.753	0.188	0.000
S26	A549 formulation	Lung epithelium; A549; in vitro	mRNA	1156	0.331	0.181	0.021
S27	A549 formulation	Lung epithelium; A549; in vitro	mRNA	436	0.325	0.114	0.046
S28	Carbonate lipid	Lung epithelium; model not reported; intratracheal	mRNA	35	-0.169	0.250	0.319
S29	Ketone 4CR	Lung epithelium; HBEC-ALI; in vitro	mRNA	28	-0.327	0.000	0.615
S30	LM 3CR	Macrophage; BDMC; in vitro	mRNA	36	0.228	0.250	0.072
S31	AGILE 3CR	Macrophage; RAW264.7; in vitro	mRNA	1100	0.115	0.173	0.174
S32	Sanofi lipid set	Muscle; mouse; intramuscular	mRNA	260	0.672	0.346	0.026
S33	Aldehyde 4CR	Muscle; mouse; intramuscular	mRNA	51	0.456	0.167	0.181
S34	Ketone 4CR	Muscle; mouse; intramuscular	mRNA	71	0.356	0.125	0.097
S35	Branched ester	Muscle; mouse; intramuscular	mRNA	18	0.276	0.000	0.236
S36	Carbonate lipid	Muscle; model not reported; intramuscular	mRNA	4	-0.200	0.000	1.000
S37	Branched ester	Spleen; mouse; intravenous	mRNA	97	0.477	0.200	0.223
S38	He aldehyde 4CR	Spleen; mouse; intravenous	mRNA	160	-0.020	0.062	0.091

Table S14. Candidate-level DeepLNP outputs for the three representative held screens. Composition gives ionizable/helper/sterol/PEG-lipid molar percentages (I/H/S/P), and the helper-lipid identity is reported separately. “Selected best” is the highest measured response among the model-ranked top decile.

Case	Candidate role	Candidate	Helper	I/H/S/P (mol%)	Measured <i>y</i>	Point output	Rank score	Model rank
C1	Highest model rank	Form 12	DOTAP	63.2/12.9/23.0/0.9	1.328	-0.120	1.427	1/157
C1	Selected and screen best	Form 22	DOPE	42.3/9.5/47.3/0.9	2.236	-0.133	1.413	11/157
C2	Highest model rank	LM_3CR_A12Iso82DC18	DOPE	35.0/16.0/46.5/2.5	0.046	-0.002	1.936	1/1080
C2	Selected and screen best	LM_3CR_A12Iso52DC18	DOPE	35.0/16.0/46.5/2.5	4.472	-0.004	1.682	3/1080
C3	Highest model rank	FO-23	DOTAP	33.8/28.5/35.2/2.5	-0.974	0.036	2.801	1/28
C3	Selected best	FO-10	DOTAP	38.8/26.4/32.5/2.4	-0.022	0.011	2.755	2/28
C3	Screen best	FO-8-propyl	DOTAP	38.0/26.7/32.9/2.4	2.517	-0.005	1.721	6/28

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S5.3. Fixed-learner partition stress test

A controlled split experiment fitted the same role-wise fingerprint Extra Trees regressor under row-random and endpoint-balanced group-disjoint five-fold partitions. Both conditions used the same feature matrix, 256 trees, minimum leaf size five, feature fraction 0.25 and seed 20260730. The matrix did not include an explicit source feature, and held-out labels were not used for tuning.

Table S15. Fixed-learner sensitivity to the partition rule. Values are fold mean $\pm$ sample standard deviation.

Metric	Row-random	Group-disjoint
RMSE	0.7762 $\pm$ 0.0240	1.0352 $\pm$ 0.0536
Pooled ρ	0.6138 $\pm$ 0.0102	0.1840 $\pm$ 0.0584
Screen ρ	0.5545 $\pm$ 0.0312	0.2287 $\pm$ 0.0973
Hit@10%	0.4480 $\pm$ 0.0642	0.2335 $\pm$ 0.0778
Norm. regret	0.1295 $\pm$ 0.0226	0.1694 $\pm$ 0.0640
Screen overlap	37.4 $\pm$ 0.5	0.0 $\pm$ 0.0

Screen overlap is the number of test-screen identities also present in training.

Row-random folds shared a mean of 37.4 of the 38 screen identities with training, whereas group-disjoint folds shared none. The row-random setting produced lower error and higher pooled and screen-local correlations (Table S15 and Figure S3). Because the partitions also alter held-out candidate distributions, this experiment estimates partition sensitivity rather than field-level leakage effects.

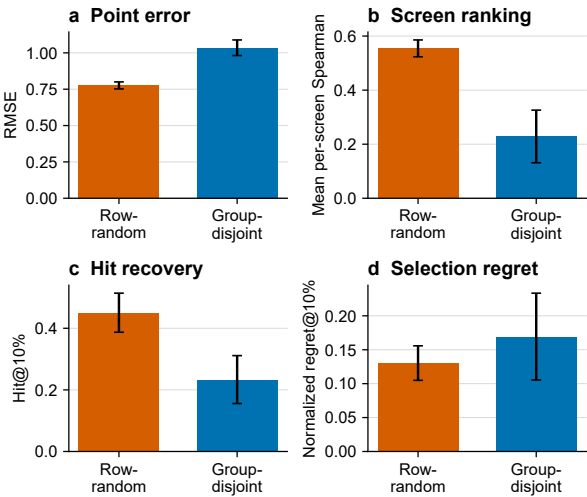


Fig. S3. Fixed-learner comparison of row-random and group-disjoint partitions. Extensive screen overlap makes row-random evaluation a within-campaign interpolation test rather than a held-screen test.

Alt text: Four paired bar charts compare RMSE, mean per-screen Spearman, top-decile hit recovery and normalized regret for row-random and group-disjoint Extra Trees models.

S6. Ablations, robustness and calibration

S6.1. Matched dual-path ablation

The ranking-path ablation changes only how the ranking score is formed. The graph-only configuration uses the graph residual in Equation S5. Direct replacement substitutes a Morgan multilayer perceptron for that residual. Direct rank-only predicts the ranking score from the Morgan pathway while leaving the point head unchanged. The selected dual residual adds the Morgan residual to the graph residual as in Equation S6. All four configurations use the same endpoint-balanced folds, training budget, sampler, losses and validation-only checkpoint procedure.

The selected A115 out-of-fold predictions cover all 12,692 delivery rows and all 38 screens exactly once, with no duplicate dataset index. Earlier objective, target, descriptor and auxiliary-task experiments are retained in the project ablation ledger; they were development experiments for the graph-only predecessor and are not pooled with the matched A115 comparison.

Three shift-robustness variants were evaluated on an earlier fixed 20% held-group development split. GroupDRO produced validation pooled Spearman 0.118 but within-screen Spearman 0.106; invariant risk minimization produced validation within-screen Spearman -0.032 and held-test Spearman -0.213 ; correlation alignment did not improve the selection

criterion [37, 38, 39]. These single-split diagnostics were not promoted into the endpoint-balanced five-fold comparison and did not alter selection of A115.

S6.2. Sensitivity to experimental-group size

Delivery screens range from four to 1,470 rows. Equal-screen aggregation represents deployment to a randomly selected screen, whereas row-size weighting gives large screens more influence. Table S17 compares these estimands and removal of the largest screen. Table S18 stratifies screens using thresholds fixed before analysis.

Excluding the largest screen changed mean screen Spearman by less than 0.002 for DeepLNP, ionizable-Morgan Extra Trees and the hybrid. Ranking estimates generally decreased with screen size, and the small-screen intervals were wide. Meta-data Extra Trees produced constant predictions in 20 screens; Tables S17 and S18 therefore summarize its 18 screens with defined Spearman correlation.

S6.3. Predictive-interval calibration

DeepLNP predicts a Gaussian mean and variance for each numeric endpoint. Table S19 compares nominal and empirical interval coverage for held-screen delivery. The 80–95% intervals were close to their nominal levels, whereas the 50% interval varied more across folds. These intervals describe the fitted

Table S16. Five-fold validation and test results for the matched ranking-path ablation. Test values are mean $\pm$ sample standard deviation.

Configuration	Validation		Held-screen test			
	RMSE	Screen ρ	RMSE	Screen ρ	Hit@10%	Norm. regret
Graph pathway only	0.9975	0.1735	1.0001 $\pm$ 0.0102	0.1017 $\pm$ 0.1483	0.1183 $\pm$ 0.0471	0.2709 $\pm$ 0.1072
Direct Morgan replaces graph rank	1.0235	0.1548	1.0470 $\pm$ 0.0714	0.1968 $\pm$ 0.0706	0.1958 $\pm$ 0.0291	0.1518 $\pm$ 0.0555
Direct Morgan rank only	1.0036	0.1791	1.0086 $\pm$ 0.0021	0.2447 $\pm$ 0.0997	0.2422 $\pm$ 0.0358	0.1234 $\pm$ 0.0728
Dual graph-Morgan residual	0.9995	0.2094	1.0029 $\pm$ 0.0033	0.2217 $\pm$ 0.0667	0.1778 $\pm$ 0.0447	0.2008 $\pm$ 0.0572

A115 (dual residual) was selected from validation performance. The higher test ranking estimate of the direct rank-only variant was observed only after selection and did not change the frozen model choice.

Table S17. Mean held-screen Spearman under alternative screen weighting. Intervals are 95% screen-bootstrap intervals.

Model	Aggregation	Estimate	95% low	95% high
DeepLNP	Equal screen weight	0.2205	0.1380	0.3040
DeepLNP	Row-count weight	0.1949	0.1339	0.2533
DeepLNP	Largest screen excluded	0.2206	0.1379	0.3078
Metadata Extra Trees	Equal screen weight	0.1854	0.0485	0.3162
Metadata Extra Trees	Row-count weight	0.1971	-0.0241	0.4381
Metadata Extra Trees	Largest screen excluded	0.1948	0.0471	0.3290
Ionizable-Morgan Extra Trees	Equal screen weight	0.2428	0.1411	0.3442
Ionizable-Morgan Extra Trees	Row-count weight	0.1843	0.1272	0.2418
Ionizable-Morgan Extra Trees	Largest screen excluded	0.2440	0.1414	0.3499
Validation-selected hybrid	Equal screen weight	0.2458	0.1430	0.3469
Validation-selected hybrid	Row-count weight	0.1901	0.1334	0.2499
Validation-selected hybrid	Largest screen excluded	0.2471	0.1429	0.3514

Table S18. Mean screen Spearman by held-out screen size. Intervals are 95% screen-bootstrap intervals.

Model	Stratum	Screens	Rows	Estimate	95% low	95% high
DeepLNP	Small (< 100)	17	795	0.2384	0.1035	0.3672
DeepLNP	Medium (100-499)	12	3,000	0.2336	0.0711	0.4102
DeepLNP	Large ($\geq$ 500)	9	8,897	0.1693	0.1078	0.2319
Metadata Extra Trees	Small (< 100)	11	578	0.1890	0.0547	0.3247
Metadata Extra Trees	Medium (100-499)	4	805	0.2553	-0.1844	0.5237
Metadata Extra Trees	Large ($\geq$ 500)	3	3,198	0.0791	-0.2771	0.4884
Ionizable-Morgan Extra Trees	Small (< 100)	17	795	0.3364	0.1333	0.5267
Ionizable-Morgan Extra Trees	Medium (100-499)	12	3,000	0.1913	0.0597	0.3145
Ionizable-Morgan Extra Trees	Large ($\geq$ 500)	9	8,897	0.1345	0.0580	0.2066
Validation-selected hybrid	Small (< 100)	17	795	0.3210	0.1195	0.5162
Validation-selected hybrid	Medium (100-499)	12	3,000	0.2118	0.0679	0.3534
Validation-selected hybrid	Large ($\geq$ 500)	9	8,897	0.1489	0.0809	0.2229

Table S19. Held-screen delivery interval coverage for A115. Values are fold mean $\pm$ sample standard deviation.

Nominal coverage	Empirical coverage
50%	0.522 $\pm$ 0.118
80%	0.803 $\pm$ 0.036
90%	0.900 $\pm$ 0.021
95%	0.949 $\pm$ 0.011
Mean absolute coverage error	0.0357 $\pm$ 0.0192
Uncertainty-absolute-error Spearman	0.094 $\pm$ 0.147

heteroscedastic head and are not conformal coverage guarantees under experimental-group shift [40].

The weak uncertainty-error correlation shows that interval calibration alone does not establish reliable sample-wise abstention. A prospective risk-coverage rule should be frozen on external calibration screens before deployment.

S6.4. Toxicity endpoint and safety limitations

Numeric toxicity supervision consists of 99 upper-bounded risk observations from two experimental groups. The values are derived only from explicit viability statements: if viability is reported as greater than $v\%$ then risk is less than $1 - v/100$. Qualitative descriptions such as “well tolerated” are not converted to labels. The retained risks lie between 0.10 and 0.15.

Table S20. Component-level evidence used to freeze the reported DeepLNP configuration.		
Component or analysis	Observed result	Consequence for DeepLNP
Graph-only ranking	Test screen ρ was 0.102; point RMSE was 1.000.	Provides the neural reference for the dual-path contribution.
Direct Morgan replacement	Test screen ρ rose to 0.197, while point RMSE worsened to 1.047.	Replacing the graph ranking residual was rejected.
Direct Morgan rank only	Test screen ρ was 0.245, but validation screen ρ was 0.179 with validation RMSE 1.004.	The test result was not used to revise the frozen selection.
Dual ranking residual	Validation screen ρ was 0.209 with RMSE 0.999; test screen ρ was 0.222.	Selected as A115 before test aggregation.
Role-wise fingerprint tree	Test RMSE 0.997 and screen ρ 0.218 did not improve jointly over the ionizable-only tree.	Additional role fingerprints are not the primary tree comparator.

Table S21. Reproducibility controls for the reported computational experiments.		
Item	Status	Required record
Dataset reconstruction	Recorded	Loader version, source-table checksums, row count and endpoint-support audit.
Chemical parsing	Recorded	Structure-resolution map, parsing failures and role-validity masks.
Split assignment	Recorded	Row identifier, group identifier, fold index and split seed for every row.
Endpoint preprocessing	Recorded	Train-only mean/standard deviation and upstream delivery-label semantics.
Model configuration	Recorded	Neural architecture, 300-tree baseline configuration, selected leaf sizes, random seeds and software environment are retained.
Checkpoint selection	Recorded	Validation objective and selected epoch for each neural fold.
Hybrid selection	Recorded	Validation-only candidate grid and independently selected point/ranking weights.
Held-out label isolation	Checked	No test label used for fitting, preprocessing, thresholding or hybrid-weight selection.
Metric implementation	Recorded	Screen-local grouping, top-decile rounding, NDCG definition and regret normalization.
Calibration	Recorded	Nominal and empirical interval coverage and uncertainty–error correlation for every neural fold.
Machine-readable results	Recorded	Fold summaries, held-screen predictions and aggregate tables used by the manuscript.
Figure provenance	Recorded	Plot script and exact machine-readable input table for every quantitative figure.

DeepLNP uses the censored Gaussian contribution in Equation S4. This loss encodes the observation direction but does not create missing dose-response information.

With no held-out toxicity-labelled group, the present evaluation provides no estimate of toxicity RMSE, ranking, calibration or transfer; the toxicity head is evaluated only as a censored-endpoint interface. A safety study would require multiple independent experiments with dose, exposure time, biological model, assay type and both toxicity and delivery measured on overlapping formulations.

S6.5. Model-selection audit

Table S20 summarizes the decisions retained after matched validation and held-screen evaluation.

S7. Reproducibility and prospective extension

Figure S4 summarizes the reproducible training and evaluation sequence. The release package prepared for the DeepLNP repository records the tested Python, PyTorch, CUDA, RDKit,

scikit-learn and plotting-library versions together with the merged-table checksum, split assignments, selected-model out-of-fold predictions and machine-readable figure inputs. Reproduction starts from immutable source tables and regenerates the loader audit, group assignments, predictions, aggregate tables and figures in that order. Changes to a source table, group key or endpoint transformation require a complete rerun with updated provenance.

S7.1. Prospective extension to complete-formulation and safety prediction

The current delivery task varies ionizable-lipid chemistry across heterogeneous screens and contains nearly complete helper-lipid structure. The positive-sterol rows use one cholesterol graph, and the PEG reagent remains graph-unresolved. Biological target and process metadata are coupled to the screening library. Group-disjoint evaluation prevents direct screen overlap but cannot identify sterol or PEG structure-activity

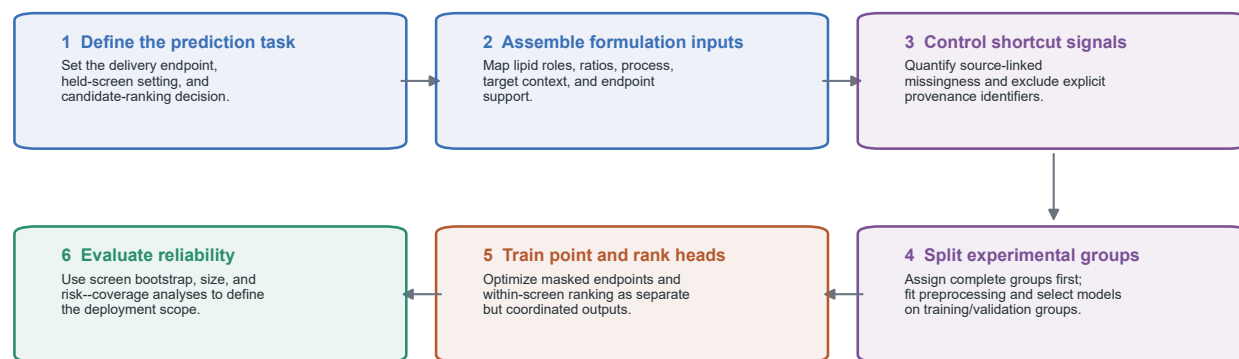


Fig. S4. Reproducible training and evaluation sequence used for DeepLNP. Input support is established before complete experimental groups are partitioned; preprocessing and model selection use training and validation groups, and point, screening and calibration outputs are evaluated separately.

Alt text: A six-stage supporting workflow links prediction-scope definition, data-support checks, group-first partitioning, task-specific metrics and model interpretation.

relationships from invariant or unresolved chemistry. Screen-standardized targets further prevent direct calibration to an absolute biological outcome.

A prospective validation should therefore be designed before selecting formulations. The experiment should cross multiple ionizable lipids with helper lipids, sterols and PEG-lipids at prespecified molar ratios, include shared controls across assay plates and retain raw, commensurate endpoints. Target and off-target delivery, viability dose response, cytokine response, serum chemistry, histopathology, particle size, PDI, encapsulation and stability should be measured on overlapping formulations with biological replication. The model, selection rule and uncertainty threshold should be frozen before opening the prospective outcomes.

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